GENERAL SCIENCE GRADE 10: TURNING EFFECT OF FORCE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 3.0: Natural Physical Science
Sub-Strand	Sub-Strand 3.1: Turning Effect of Force
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Moments of a force at a point – Moments of anti-parallel forces – Applications of the turning effect of force in real life
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine moments of a force at a point, b) calculate the moments of anti-parallel forces, c) apply the turning effect of force in real-life situations, d) appreciate the importance of the turning effect of force in everyday life.
Core Competencies	– Critical Thinking and Problem Solving: as the learner solves numerical problems involving both moments of a force and moments of anti-parallel forces and identifies areas where moments of a force is used in day-to-day life. – Digital Literacy: as the learner interacts with digital technology when researching on moments of a force at a point and moments of anti-parallel forces.
Core Values	– Integrity: the learner displays honesty while using digital devices to search for information on moments of a force at a point and moments of anti-parallel forces. – Respect: the learner respects others' opinions while discussing with peers on the examples involving turning effect of force in daily life.
Science & Engineering Practices	– Asking Questions and Defining Problems: as learners generate questions to explain why a panga or jembe handle length affects cutting/digging effort, framing the driving question for the unit. – Planning and Carrying Out Investigations: as learners design and carry out the metre-rule balancing experiment, controlling variables (distance, load) to determine the moment of a force at a point. – Analysing and Interpreting Data: as learners record force and distance measurements, compute moments, and identify patterns linking moment magnitude to lever arm length. – Using Mathematics and Computational Thinking: as learners solve numerical problems involving moments of a force and moments of anti-parallel forces using the principle of moments ($F \times d = F \times d$). – Constructing Explanations: as learners explain real-life applications — wheelbarrows on Nairobi construction sites, seesaws at school playgrounds, jembe handles on Rift Valley farms — using the moment equation.

	– Obtaining, Evaluating, and Communicating Information: as learners research turning-effect applications using digital devices and print media, then present findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Self-Awareness): as the learner demonstrates the principle of moments of force by balancing a metre rule, developing awareness of how their own body uses turning effects (e.g., lifting water jerricans, opening doors).
Career Connections	– Civil and Structural Engineering: understanding moments and turning effects is fundamental to designing bridges, buildings, and infrastructure projects such as those along the Nairobi Expressway. – Agriculture and Farm Mechanisation: farmers in the Rift Valley and Kisumu region use levers (jembes, ploughs, wheelbarrows) daily; agronomists and farm-equipment designers rely on moment principles. – Physiotherapy and Sports Science: Kenyan athletes and physiotherapists apply torque concepts when analysing joint mechanics and designing rehabilitation programmes. – Architecture and Construction Technology: architects and builders designing structures such as cantilever roofs on Mombasa beach resorts apply turning-effect principles for structural stability. – Mechanical Engineering and Automotive Technology: mechanics working on matatu and truck engines use spanner torque calculations directly derived from moments of a force.
Focus for Lessons	This unit develops learners' understanding of how forces produce rotational effects — moments — and how the principle of moments governs balance and mechanical advantage in tools and structures. Using a phenomenon-first approach anchored in the everyday experience of Kenyan learners, lessons progress from qualitative prediction and hands-on investigation with a metre rule, through quantitative problem-solving with anti-parallel forces, to critical evaluation of real-life applications including seesaws, wheelbarrows, jembes, door handles, and construction cranes. The unit embeds NGSS Science and Engineering Practices and KICD core competencies throughout, with a consistent Kenyan context.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does where you push, pull, or grip something determine how easy or hard it is to make it turn — and how do engineers and farmers in Kenya use this to design better tools? KICD KEY INQUIRY QUESTION: 1. How is the turning effect of force used in our daily life?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Jembe Handle Puzzle: A farmer in Kericho uses a long-handled jembe (hoe) to break hard soil with seemingly little effort, yet when the same farmer grips the jembe close to the metal blade, even soft soil becomes exhausting to dig. Students observe a short video clip (or teacher demonstration) of two learners using jembes of identical blade weight but different handle lengths to dig the same patch of school-garden soil: the learner with the longer handle finishes quickly and easily, while the learner with the shorter handle struggles and tires rapidly. This is puzzling because both jembes have the same blade (same force of gravity on the tool) and the same person is applying effort — so why does handle length change how hard the work feels? Students are further shown that simply sliding their grip further from the blade pivot makes the task easier, even though nothing else changes. The phenomenon is compelling because it is universally familiar to Kenyan learners from home and farm life, yet the underlying rotational mechanics — that effectiveness depends on both the size of the force AND its distance from the pivot — is not intuitively obvious and demands scientific explanation.
Storyline Thread	Lesson 1 — PREDICT (Anchoring Phenomenon Introduction): Teacher presents the Jembe Handle Puzzle via demonstration or short video clip. Learners observe the two-jembe scenario and individually write predictions: "Why does the longer-handled jembe make digging easier?" Class shares predictions; teacher records all ideas on the Driving Question Board (DQB) without confirming or correcting. Learners are introduced to the driving question and begin an initial model sketch showing "where the force goes" when using a long vs. short jembe handle.

	Lesson 2 — OBSERVE/INVESTIGATE (Moments of a Force at a Point — Metre Rule Experiment): Learners carry out the metre-rule balancing experiment: a ruler balanced on a pencil pivot, with known masses hung at measured distances on each side. Groups collect force (weight) and distance data, record in tables, and compute F × d for each side. Patterns emerge: balance occurs when $F_1 \times d_1 = F_2 \times d_2$. Learners connect results back to the jembe — the pivot is the blade's contact point with the soil, and the moment is effort × handle length. Lesson 3 — EXPLAIN (Formalising the Moment Equation): Teacher formally introduces the definition of the moment of a force ($M = F \times d$, unit: Newton-metre, Nm), the principle of moments, and conditions for rotational equilibrium. Learners use digital devices or print media to research moments of a force and watch animations/simulations. Worked examples use Kenyan contexts: a market vendor's weighing balance in Gikomba market, a child on a seesaw in Uhuru Park. Learners solve structured numerical problems (increasing complexity) and revisit their initial predictions, annotating what was right or wrong. Lesson 4 — OBSERVE/INVESTIGATE (Anti-Parallel Forces and Couples): Learners are introduced to anti-parallel forces through the supporting phenomenon of the wheelbarrow and the door-handle. Teacher demonstrates (or learners investigate) systems where two parallel but opposite forces act on an object — a couple. Learners measure and calculate the resultant turning effect of anti-parallel force pairs using the metre rule or a suspended beam with two force meters pulling in opposite directions. They establish that the net moment of a couple = F × perpendicular distance between the forces. Lesson 5 — EXPLAIN (Calculating Moments of Anti-Parallel Forces — Numerical Problems): Learners consolidate understanding of anti-parallel forces through structured problem-solving. Problems are drawn from Kenyan contexts: a cart axle pulled by two oxen in the Rift Valley, a steering wheel turned by two hands, a matatu door handle. Learners work in pairs, peer-checking calculations. Teacher uses cold-calling (minimum 4 learners per problem set) and 30-second wait time before accepting answers. Common misconceptions about direction and sign of moments are explicitly addressed. Lesson 6 — OBSERVE/APPLY (Walk-Around and Real-Life Application Mapping): Learners walk around the school compound and neighbouring area to identify and document at least five real-life examples of turning effect of force (gates, flagpoles, tap handles, scissor hinges, window stays). They photograph or sketch each example, label the pivot, effort, load, and moment arm, and calculate estimated moments where possible. Back in class, groups present findings; the class watches short video clips on cranes, pliers, and bicycle brakes — all using turning-effect principles in Kenyan infrastructure and transport contexts. Lesson 7 — DQB UPDATE AND MODEL BUILDING: Learners return to the Driving Question Board and systematically answer or revise each question posted in Lesson 1. They update their initial jembe model (from Lesson 1) into a fully labelled scientific diagram showing pivot point, effort force, load force, perpendicular distances, and calculated moments. Groups compare their revised models with peers and discuss how the same principle explains the door handle, seesaw, and wheelbarrow supporting phenomena. Teacher facilitates a whole-class sensemaking discussion linking all phenomena to a single underlying explanation: the moment equation. Lesson 8 — CONSOLIDATION, ASSESSMENT AND REFLECTION: Learners complete an individual written task involving unseen numerical problems on moments and anti-parallel forces using a new Kenyan context (e.g., a fisherman at Lake Victoria using a boat oar as a lever). Teacher administers an oral/practical component: each learner demonstrates the principle of moments by balancing a metre rule with given masses and explains the result. Lesson closes with a reflection journal entry: "How has your understanding of turning effect of force changed since Lesson 1, and where will you notice it this week at home or on the farm?"

Supporting Phenomena	– Seesaw Imbalance at a School Playground: Two learners of different body masses sit on opposite ends of a seesaw at a Nairobi primary school. The heavier learner's side always drops. But when the heavier learner slides closer to the pivot, the seesaw can be made to balance — even though nobody changed in mass. Why does position matter? – The Door-Handle Position Puzzle: A heavy school gate in Kisumu swings open easily when pushed near the outer edge, but requires enormous effort when pushed near the hinges. Both pushes use the same gate and the same gravity — what makes the outer-edge push so much more effective? – Wheelbarrow on a Construction Site: A fundi (builder) in Mombasa loads heavy cement blocks into a wheelbarrow and lifts the handles to move the load. The single wheel acts as a pivot. Learners notice that placing the load above the wheel (close to the pivot) is much harder to lift than placing it further back — the opposite of what they might expect. This sets up anti-parallel forces and distributed loads. – Spanner and Bolt: A mechanic tightening bolts on a matatu engine in Nairobi uses a long spanner rather than a short one, even though both fit the same bolt. Adding a metal pipe to extend the spanner handle further reduces the effort needed. Students predict which spanner will loosen a tight bolt more easily, then investigate why.

LESSON 1 (40 minutes): The Jembe Handle Puzzle — Launching the Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce the anchoring phenomenon — the Jembe Handle Puzzle — and activate learners' prior knowledge about force and turning, so they can generate testable predictions and begin building an explanatory model for why handle length affects the ease of digging.
Knowledge	Learners will know that the effectiveness of a turning force depends on both the size of the force applied and the distance from the pivot point at which it is applied (qualitative introduction only at this stage).
Skills	Learners will be able to: (i) observe and describe a phenomenon carefully; (ii) formulate written predictions using evidence-based reasoning; (iii) create an initial diagrammatic model showing force application on a jembe handle; (iv) pose investigable questions for the Driving Question Board.
Attitudes	Learners appreciate that everyday Kenyan farming tools embody scientific principles, and develop curiosity and intellectual humility by recognising that their initial predictions may need revision as evidence accumulates.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	This is the opening lesson of the sub-strand. It launches the anchoring phenomenon (the Jembe Handle Puzzle) that will anchor all subsequent lessons on moments of a force and anti-parallel forces. Learners do NOT receive explanations today — the lesson is designed to surface prior conceptions, generate wonder, and open the Driving Question Board. Every later lesson will return to this jembe context to deepen the explanation.
Safety Notes	If a physical jembe is used for demonstration, the teacher must designate a clear safety zone of at least 1.5 m around the demonstrator. No learner should handle the jembe without explicit permission. Ensure the demonstration area (school garden or open ground) is free of stones and other learners before swinging. If a video is used instead, no physical safety concerns arise.

B. LESSON OVERVIEW
Lesson 1 opens the entire Turning Effect of Force unit by presenting the Jembe Handle Puzzle as the anchoring phenomenon. The teacher shows a short video clip (or live demonstration) of two learners using jembes with identical blade weights but different handle lengths to dig the same patch of school-garden soil in Kericho. Learners individually write predictions, share them with the class, and see all ideas recorded — unconfirmed — on the Driving Question Board. The lesson closes with learners sketching an initial explanatory model of "where the force goes" when gripping a long versus short jembe handle. No scientific vocabulary (moment, torque, pivot) is formally introduced yet; this lesson is purely about noticing, wondering, and predicting.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Equation with the variable in the	Learners watch a 2–3 minute video clip (or observe a live teacher demonstration) of two learners — Amina and Juma —	BEFORE VIDEO: Teacher says exactly — 'Before I show you anything, I want you to think about the last time you or someone in	Strategy: INDIVIDUAL WRITTEN PREDICTION BEFORE INSTRUCTION (Elicit-Before-Explain). Rationale: Research	Observable indicator: Every learner has written at least 2 sentences on their PREDICT sheet within 4 minutes. Teacher

denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	digging the same patch of school-garden soil in Kericho. Amina uses a jembe with a long handle (≈90 cm from blade pivot to grip); Juma uses a jembe with a short handle (≈30 cm from blade pivot to grip). Both jembes have identical metal blades. Amina finishes her row quickly and looks relaxed; Juma struggles, tires rapidly, and barely breaks the soil. Learners then individually complete a structured PREDICT sheet (printed or in exercise books) with three prompts: (1) 'What did you notice?' (2) 'Why do you think Amina found it easier than Juma — write your best explanation in 2–3 sentences.' (3) 'What information would you need to test your explanation?' Learners write silently for 4 minutes with no peer discussion — predictions must be their own uninfluenced ideas.	your family used a jembe. Where did you grip it? Did it matter?' Allow 15 seconds WAIT TIME for private thinking. Then: 'You are about to watch a short clip. Your job is to be a scientist — notice everything, judge nothing yet.' Play the video or conduct the live demonstration. DURING VIDEO: Teacher is silent — do not narrate or explain. AFTER VIDEO (immediate): Teacher says — 'In your exercise book, write the heading: MY PREDICTION — Lesson 1. Now write answers to the three questions on the board. You have 4 minutes. Do not talk to your neighbour yet — I want YOUR thinking, not your friend's.' Teacher circulates, reading over shoulders, and uses a tally on a clipboard to note three categories of prediction emerging: (A) predictions about force size, (B) predictions about distance/position, (C) other. Teacher does NOT correct or affirm any prediction. After 4 minutes, teacher says — 'Pens down. Now, cold-call: I will ask five of you to read your prediction aloud — anyone can be called.' Cold-call 5 learners deliberately choosing a spread: 1 from category A, 1 from B, 1 from C, 1 who seems uncertain, 1 high-confidence writer. After each reading, teacher says only: 'Thank you. Who predicted something similar? Hands up.' Count hands aloud: 'I see eight hands — noted.' Record each prediction verbatim on the board under the heading WHAT WE THINK SO FAR.	consistently shows that committing a prediction to writing before discussion activates prior knowledge, surfaces misconceptions for the teacher to monitor, and creates a personal benchmark learners can revisit and revise. By requiring written predictions before any peer talk, the teacher ensures every learner engages cognitively — not just the most vocal. The teacher's clipboard tally of prediction categories enables targeted questioning in later phases.	uses clipboard tally to identify: (i) learners who write only 'the longer one is better' with no causal reasoning (need scaffolding in Phase 2); (ii) learners who spontaneously mention distance or position from the blade (strong prior knowledge — can be used as peer resources later). Teacher records initials of 3–4 learners with undeveloped predictions to check in with during Phase 3.
Observe Phase 📹 **VIDEO:**	The teacher replays the video (or repeats the demonstration) — this time learners use an	Before second viewing: 'This time, you are collecting evidence — like a detective. Use your observation	Strategy: STRUCTURED OBSERVATION WITH EVIDENCE GUIDE + PHYSICAL ANALOGY.	Observable indicator: Learner DIFFERENT columns should include at least one reference to

Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	OBSERVATION GUIDE to collect structured evidence. The guide has four columns: (1) What I see happening with Amina's long-handled jembe; (2) What I see happening with Juma's short-handled jembe; (3) What is the SAME about both jembes/situations; (4) What is DIFFERENT. Learners complete the guide individually during the second viewing (3 minutes), then share in pairs for 2 minutes using the sentence stem: 'I noticed ______ which made me think ______.' After pair-share, the teacher conducts a whole-class share-out: cold-calls 4 pairs — 'Tell me one thing from your SAME column' and 'Tell me one thing from your DIFFERENT column.' Teacher records all observations on a two-column T-chart on the board: SAME | DIFFERENT. Teacher then introduces a physical analogy: holds up a long ruler and a short pencil and asks — 'If I push at the far end of this ruler, what happens? What if I push right next to my finger?' Learners try this with their own rulers at their desks for 60 seconds — no formal instruction, just play.	guide. I will pause the video at 1 minute 10 seconds — use that pause to catch up on your writing.' Play video; pause at 1:10; allow 20 seconds writing time; resume. After video: 'Turn to the person next to you. You have exactly 2 minutes — use the sentence stem on the board: I noticed _____ which made me think _____. Go.' Teacher circulates and listens actively; notes 2 observations that surprise or that contain nascent scientific thinking. After 2 minutes: 'Time. Let us share. I will cold-call four pairs.' Cold-call pairs (not individuals — reduces anxiety). After recording SAME/DIFFERENT T-chart: 'Now — pick up your ruler or a pencil. Push at the far end. Now push right next to your pinching fingers. What do you feel? You have 60 seconds — just notice.' After 60 seconds: 'One word — how does pushing far away feel compared to pushing close? Shout it out — all at once on three. One, two, three—' (whole-class simultaneous response). Record the dominant words (e.g., 'easier', 'stronger', 'less effort') on the board beside the T-chart without explanation.	Rationale: The two-column observation guide trains learners to distinguish observations from inferences and to identify variables (same vs. different) — a foundational NGSS Science and Engineering Practice (SEP 3: Planning and Carrying Out Investigations). The ruler analogy bridges the farm context to a classroom-manipulable object, making the phenomenon tangible for learners who may not have used a jembe personally. The simultaneous whole-class shout-out lowers affective risk for the word-generation step.	handle length, grip position, or distance from the blade. Teacher scans guides during pair-share. Red flag: if fewer than 60% of learners mention distance/position in their DIFFERENT column, the teacher inserts an additional prompt — 'Look specifically at where each person is holding the jembe relative to the metal blade — what is different there?' This re-focuses observation without giving away the explanation.
Explain Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic	Learners return to their original PREDICT sheet and use a coloured pen to annotate: 'What would I change or add to my prediction now that I have observed more carefully?' (2 minutes silent writing). Then the class engages in a structured THINK-PAIR-SHARE around the question: 'What is it about where you hold the jembe that matters — and why?' Pairs share with another pair (groups of 4) for 3 minutes; each group of 4 must agree on	Distribute coloured pens (or ask learners to use a different ink colour). 'Go back to your prediction. Do NOT cross anything out — use your coloured pen to add or change. You have 2 minutes.' After 2 minutes: 'Now share with your partner — then join with the pair behind you. Your group of four must write ONE agreed sentence on this strip of paper: We think the distance from the blade pivot matters because ___. You have 3 minutes.'	Strategy: PREDICT-OBSERVE-EXPLAIN (POE) CYCLE with ANNOTATED PREDICTION REVISION. Rationale: Asking learners to annotate rather than replace their original prediction makes their thinking visible and honours the iterative nature of scientific reasoning (NGSS SEP 6: Constructing Explanations). The finger-vote is a low-stakes simultaneous response that gives the teacher instant whole-class data on conceptual readiness	Observable indicator: The group agreed-sentences should move beyond 'the longer one is better' toward causal language referencing position or distance. Teacher listens during group work for groups still using purely force-size language ('Amina pushes harder') — these groups need the prompt about the ruler activity. Finger-vote tally: if more than 25% of learners vote for grip position A or B (blade end), this indicates a significant misconception to

Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	one shared sentence that starts: 'We think the distance from the blade pivot matters because ___.' Each group writes their agreed sentence on a sticky note (or torn paper strip). Teacher collects all sentences, reads them aloud, and groups them on the board into clusters by theme — without yet confirming any. Teacher then shows a labelled diagram of a jembe (drawn on the board or projected) indicating three points: (A) the metal blade tip touching soil (the pivot/fulcrum), (B) the middle of the handle, (C) the far end of the handle. Teacher asks: 'If you were Amina, which point would you grip — A, B, or C? Show me with your fingers. Hold up one, two, or three fingers — on my count.' Simultaneous finger-vote. Teacher tallies. Then asks: 'Juma gripped near A — why does that make it harder? Write one sentence — you have 90 seconds.'	Circulate; when a group is stuck, prompt with: 'Think about the ruler activity — what changed when you pushed further away? Only the position of your push changed — what else might change?' Collect paper strips. Read each aloud with no comment other than 'Interesting' or 'Thank you.' Cluster on board. Draw the jembe diagram. 'Three points — A is the blade, C is the far end of the handle. Where does Amina grip? Show me fingers — 1 for A, 2 for B, 3 for C. On three — one, two, three.' Tally finger votes aloud. Then: 'Write one sentence explaining why gripping near A — near the blade — makes it harder. 90 seconds. Go.' Cold-call 3 learners (choose 1 from the 3–4 flagged in Phase 1 as needing support — to check growth). After each: 'Does anyone want to add to that explanation?' Allow 10 seconds WAIT TIME before accepting additions.	without singling out individuals. Grouping sticky-note sentences into thematic clusters on the board is a teacher-led data analysis move that models how scientists look for patterns across multiple explanations.	address explicitly in Lesson 2 when the pivot concept is formalised. Teacher notes the tally in their register.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12	The teacher formally introduces the Driving Question Board — a dedicated space on the classroom wall (or a large sheet of paper) that will remain visible for the entire sub-strand. The teacher writes the Driving Question at the top in large letters: 'WHY does where you push, pull, or grip something determine how easy or hard it is to make it turn — and how do engineers and farmers in Kenya use this to design better tools?' Learners are each given two small cards (or paper strips): Card 1 — 'Something I notice or wonder from the jembe puzzle'; Card 2 — 'A question I have that I do not yet know how to answer.' Learners write individually (2 minutes), then	Distribute two cards per learner. 'Card 1: something you notice or wonder. Card 2: a question — something you genuinely do not know the answer to yet. No question is too simple. You have 2 minutes.' After writing: 'Come up row by row — stick Card 1 in the WONDERINGS zone, Card 2 in the QUESTIONS zone.' Teacher stands at the DQB and reads cards aloud as learners sit down: 'I am seeing questions about — ' (cluster live): 'why position matters, what happens with other tools like a wheelbarrow or a door, whether heavier people push harder, how farmers in Rift Valley build their tools.' After clustering: 'These are exactly the kinds of questions	Strategy: DRIVING QUESTION BOARD (DQB) as a KNOWLEDGE-TRACKING ARTEFACT. Rationale: The DQB is a core phenomenon-anchored sensemaking tool drawn from the Next Generation Science Standards storyline approach. By making learner questions public and persistent, it: (i) validates curiosity as a scientific practice (NGSS SEP 1: Asking Questions); (ii) creates accountability — learners can see which questions they have answered in subsequent lessons; (iii) gives the teacher a real-time map of the class's conceptual landscape. Clustering questions by theme models the scientific practice of organising	Observable indicator: Every learner produces two cards. Quality indicator — Card 2 (question) should be phrased as a genuine question (contains a question word: why, how, what, does). Red flag: cards that are statements rather than questions ('I think long handles are better') suggest the learner has not yet shifted from assertion to inquiry mode — teacher notes these learners for targeted questioning in Lesson 2. Teacher also checks whether any Card 2 questions anticipate the quantitative relationship (e.g., 'Does the length matter by a specific amount?') — these learners may be ready for extension numerical problems in

🔍 Search ARES for similar readings	come up and stick both cards on the DQB in designated zones: WONDERINGS | QUESTIONS. Teacher reads out 5–6 cards, grouping them live on the board: questions about force, questions about distance, questions about other tools, questions about the soil. The teacher explicitly tells the class: 'Every lesson in this sub-strand, we will come back to this board. Some of your questions will be answered by experiments. Others you will answer by calculation. By the last lesson, this board should tell the story of everything you have figured out.'	scientists ask. Our driving question ties all of these together.' Point to the driving question written at the top. Read it aloud twice — ask the class to read it chorally once. Then: 'This board belongs to you. I am the keeper — but the ideas are yours. Let us find out together.' Teacher photographs the DQB with a phone or tablet for the ARES platform record.	data for pattern-finding.	Lesson 3.
Model Building Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners create their INITIAL MODEL — a diagrammatic sketch in their exercise books under the heading 'My First Model — Lesson 1.' The model must show: (a) a labelled sketch of a jembe, clearly marking the metal blade, the handle, and the point where the farmer grips; (b) arrows showing where the farmer's force is applied and in which direction; (c) a dot or circle marking what they think the pivot point is (where the jembe rotates); (d) a written sentence: 'I think the longer handle makes digging easier because ___.' Learners work individually for 4 minutes, then share their model with a partner for 1 minute using the prompt: 'Point to where you drew the force and where you drew the pivot — are they in the same place as mine?' The teacher collects or photographs 4–5 models (choosing variety — detailed, simple, unconventional) to display anonymised at the start of Lesson 2 as comparison points for model revision.	Write on the board: 'MY FIRST MODEL — Lesson 1. Draw a jembe. Label: blade, handle, grip position. Draw arrows for force. Mark the pivot with a dot. Write one sentence.' Say aloud: 'Your model does not have to be perfect — it has to show your thinking right now. Scientists draw rough models all the time. Dr. Mwangi Gitahi, the Kenyan engineer who designs agricultural tools for small-scale farmers in the Rift Valley, started with rough sketches just like these. You have 4 minutes.' Circulate; after 2 minutes, prompt any learner with a blank page: 'Just draw the stick shape of a jembe — now where does the farmer hold it? Put a hand there. Now which way does the soil push back — draw that arrow.' After 4 minutes: 'Show your partner your model. Point to your force arrow. Point to your pivot dot. Do you agree on where the pivot is? You have 1 minute.' After 1 minute: 'I am going to collect five models to share next lesson — anonymously. Who is happy for	Strategy: INITIAL MODEL CONSTRUCTION (NGSS SEP 2: Developing and Using Models). Rationale: Asking learners to commit a diagrammatic model to paper before any formal instruction creates a concrete, revisable artefact of their current mental model. The act of drawing — especially placing force arrows and a pivot dot — forces precision that verbal description does not: learners must decide specific spatial relationships. Partner comparison of pivot placement is a deliberate move to surface the most common early misconception (that the pivot is anywhere along the handle, rather than at the blade-soil contact point). Collecting and photographing models gives the teacher formative data on the distribution of mental models across the class, and provides anchor images for model-revision discussions in Lessons 2–5.	Observable indicator: Every learner sketch contains at least (i) a jembe outline, (ii) one force arrow, (iii) a marked pivot location, and (iv) one written sentence. Quality spectrum — Teacher looks for three levels: Level 1: force arrow placed anywhere, pivot unmarked or placed at grip (common misconception — pivot is where you hold it); Level 2: force arrow at grip, pivot placed at blade end (partially correct); Level 3: force arrow at grip, pivot correctly at blade-soil contact, sentence references distance (most sophisticated). Teacher records the distribution of these three levels to plan differentiated entry points for Lesson 2. Models at Level 1 need the pivot concept introduced concretely with a physical pivot demonstration at the start of Lesson 2.

		me to photograph theirs?' Collect 4–5 with consent. Close: 'Your model is your current best thinking. Every lesson, you will update it. By the end of this sub-strand, you will be amazed at how much more precise and powerful your model becomes — and you will be able to explain exactly why Amina's jembe works and how to design the best handle for any farming tool in Kenya.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal:

1. PREDICTION QUALITY: What proportion of learners' written predictions referenced the position or distance of the grip (rather than only force size or personal strength)? If fewer than 40% mentioned position/distance, plan to replay the ruler analogy at the start of Lesson 2 with more explicit guidance.
2. DQB RICHNESS: Count the number of distinct question themes on the DQB. A healthy DQB for Lesson 1 should have at least 4 distinct themes (e.g., force, distance, other tools, farmer/engineering application). If fewer than 3 themes emerged, the class may need more open-ended prompting — consider adding a 'parking lot' card in Lesson 2 where you deliberately model asking a wondering question yourself.
3. MODEL SOPHISTICATION: Using your photographs of the 4–5 collected models, classify them by the three levels described in Phase 5 formative assessment. If more than 50% are at Level 1 (pivot at grip or unmarked), begin Lesson 2 with a physical pivot demonstration using a metre rule balanced on a pencil before introducing any vocabulary.
4. ENGAGEMENT: Which learners did not contribute a card to the DQB or left cards blank? Visit these learners individually in the first 5 minutes of Lesson 2 to verbally gather their wonder/question and add it to the board on their behalf — this signals that every voice belongs on the board.
5. KENYAN CONTEXT CONNECTION: Did the jembe phenomenon genuinely resonate — did learners spontaneously reference their own home or farm experiences? Note any additional Kenyan tool contexts learners raised (e.g., wheelbarrow, panga, mkokoteni cart, water pump handle) — these can be woven into Lessons 3 and 4 as application examples.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two learners — Amina and Juma — digging school-garden soil in Kericho using jembes with identical blade weights but different handle lengths. Amina (long handle ≈90 cm) finished quickly and easily; Juma (short handle ≈30 cm) struggled and tired rapidly. A ruler push-activity showed that pushing at the far end feels easier than pushing close to the pivot finger.
What did I learn?	We have not yet learned the scientific explanation — we are still in the prediction stage. We noticed that handle length and grip position seem to matter. We generated predictions and questions. We created our first model sketch of force arrows and a pivot on a jembe. Our Driving Question Board is now open and holds all our current wonderings and questions.
How does this explain the phenomenon?	Nothing has been formally explained yet — this is intentional. Our best current prediction is that gripping further from the metal blade makes digging easier, but we do not yet know WHY in scientific terms. We will investigate this in Lesson 2 using a metre rule and hanging masses to find the pattern.

LESSON 2 (80 minutes (2 × 40-minute lessons)): Moments of a Force at a Point — The Metre Rule Balancing Investigation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners carry out a hands-on metre-rule balancing experiment to gather quantitative evidence for the Principle of Moments, connecting measured F × d values directly to why handle length matters on the jembe.
Knowledge	Learners determine that the moment of a force at a point equals force multiplied by perpendicular distance from the pivot (M = F × d), and that rotational equilibrium occurs when the sum of clockwise moments equals the sum of anticlockwise moments.
Skills	Learners measure distances with a metre rule, calculate weights from known masses (W = mg), record data in structured tables, compute F × d products, identify patterns across multiple trials, and construct an evidence-based claim.
Attitudes	Learners appreciate that precise measurement and honest data recording are essential for trustworthy scientific conclusions, and demonstrate integrity when results do not match predictions.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	In Lesson 1 learners were puzzled by the jembe phenomenon and made initial predictions. Lesson 2 is the OBSERVE/INVESTIGATE phase: learners now gather real quantitative data using a metre-rule pivot experiment, discover the F × d pattern, and use it to begin explaining why the long-handled jembe in Kericho works so much better — advancing the driving question from 'I wonder why' to 'I have evidence that…'.
Safety Notes	Masses must be secured with loops of string — no free-hanging masses over learners' feet. The pencil pivot must be held steady or taped to a flat desk surface. Learners must not lean over the suspended rule. Broken metre rules with splinters must be reported to the teacher immediately and replaced.

B. LESSON OVERVIEW
Learners open by revisiting their Lesson 1 jembe predictions on the Driving Question Board. They then conduct a structured metre-rule balancing experiment in groups of four: a wooden metre rule is balanced on a pencil pivot at the 50 cm mark, and groups systematically hang known masses (using 100 g, 200 g, and 500 g slotted masses) at measured distances on each side, recording force (weight in Newtons, using W = mg with g = 10 N/kg) and distance (in metres) in a data table. Groups compute F × d for each side across at least five trial configurations. A whole-class data share reveals the invariant pattern: balance always occurs when $F_1 \times d_1 = F_2 \times d_2$. Learners name this the Principle of Moments, then immediately re-examine the jembe: the soil contact point is the pivot, effort is the applied force, and handle length is the perpendicular distance — making the moment (turning effect) larger with a longer handle for the same effort force. The lesson closes with a DQB update and an initial revision of learners' predictive models.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Equation with the](#)	Learners individually re-read their Lesson 1 written predictions about the jembe (kept in their science	Display the two jembe diagrams on the board (drawn or projected). Say verbatim: 'You all saw the	Prediction Journaling with Public Commitment — by writing and then publicly sharing predictions	Teacher scans journals during the 3-minute writing window and notes: (a) learners who write only

variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	journals). The teacher projects two jembe diagrams side by side — one with a 1.2 m handle, one with a 0.3 m handle — and poses a focused quantitative challenge: 'If I push DOWN with exactly 50 N of effort on each jembe, which one turns the blade more forcefully? Can you put a number on it?' Learners write a prediction in their journals, including a guess about what quantity they should calculate to compare the two scenarios. Groups of four briefly share their predictions (2 minutes), then a spokesperson from each group states their group's best idea to the class.	Kericho farmer puzzle in our last lesson. Today we are going to stop guessing and start measuring. Before we touch any equipment, I want you to commit to a prediction in your journal — write it now, even if you are unsure.' Allow 3 minutes of silent individual writing. After writing, cold-call FIVE different learners (across gender and seating zones) to share predictions — do NOT evaluate answers yet, simply record key phrases on the board under the heading 'Our Predictions.' Say: 'Hold onto these — by the end of today you will know which of these ideas the data supports.' Apply 12-second WAIT TIME after each cold-call before moving to the next learner.	before data collection, learners activate prior knowledge, create cognitive dissonance targets, and generate personal questions that drive attentive observation during the experiment. Recording predictions verbatim on the board makes them falsifiable and creates accountability.	qualitative statements ('longer handle is easier') — these learners need prompting to think about numbers; (b) learners who spontaneously write a ratio or product — these are ready to lead data analysis. Teacher makes a mental note of at least two learners from each category for targeted questioning during the experiment phase.
Observe Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four collect apparatus: one wooden metre rule, one pencil (pivot), four 100 g slotted masses, two 200 g slotted masses, one 500 g slotted mass, string loops pre-cut to 5 cm, and a printed data-recording table (or hand-drawn in journal). Step 1 — Groups balance the bare metre rule on the pencil at exactly the 50 cm mark and confirm balance (zero-load check). Step 2 — One learner hangs a 200 g mass (weight = 2 N) at 30 cm from the pivot on the LEFT side. The group's task: find a position on the RIGHT side where a 100 g mass (1 N) balances it. They record both distances and both weights. Step 3 — Groups complete at least FOUR more trial configurations of their own choosing (varying both masses and distances), always recording: Mass Left (g), Weight Left F_1 (N), Distance Left d_1 (cm), Mass Right (g), Weight Right F_2 (N), Distance	Distribute apparatus efficiently (pre-packed in labelled trays — one tray per group). Demonstrate the zero-load balance check as a whole class before groups begin: 'Watch — I place the rule at exactly 50 cm on the pencil. It must balance with nothing hanging. If it does not, the rule has an uneven mass distribution and we record its natural balance point instead.' Circulate continuously during the experiment. At the 10-minute mark, visit each group and ask: 'What does $F_1 \times d_1$ equal for your first trial? What does $F_2 \times d_2$ equal? What do you notice?' Apply 15-second WAIT TIME before prompting. At the 18-minute mark, if any group has fewer than three completed trials, say: 'You need at least five data rows to see a pattern — keep going, focus.' If a group finds an unexpected result ($F_1 \times d_1 \neq F_2 \times d_2$), say: 'Excellent — don't erase that. Measure again	Data-Table Pattern Detection — learners generate their own numbers across varied configurations, making the F×d invariance an empirical discovery rather than a received fact. The five-trial minimum ensures the pattern is not coincidental. The 'We noticed…' sentence forces groups to articulate the pattern in their own words before the teacher names it formally — a key NGSS Science and Engineering Practice (Analysing and Interpreting Data; Constructing Explanations).	Teacher uses a clipboard checklist to verify each group has: (1) at least five completed data rows with units, (2) both F×d columns calculated, (3) a written 'We noticed…' claim. Any group missing items 1–3 by the 28-minute mark receives a direct prompt. Teacher also listens for correct use of units (N·cm or N·m) — if any group records F×d without units, address this explicitly in the whole-class share.

	Right d_2 (cm), $F_1 \times d_1$ (N·cm), $F_2 \times d_2$ (N·cm). Step 4 — Each group computes all F×d products and looks for a pattern. Step 5 — Groups write a one-sentence 'We noticed…' claim in their journals before the class share.	carefully and record both results. Real scientists keep anomalies.' At the 28-minute mark, call the class to attention: 'Finish your current trial and write your one-sentence claim NOW — you have 2 minutes.' Cold-call THREE groups to read their claim aloud before transitioning to Phase 3.		
Explain Phase 📹 **VIDEO:** [Equation with the variable in the denominator]() Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos]() 📖 **HTML:** [Moment of Inertia (Flexbook)]() Source: CK-12 🔍 [Search ARES for similar readings]()	Whole-class data share: a scribe (rotating student volunteer) records each group's $F_1 \times d_1$ and $F_2 \times d_2$ values from two representative trials on the class data board. Learners observe that across all groups, $F_1 \times d_1 \approx F_2 \times d_2$ for every balanced configuration. The teacher formally introduces the term MOMENT OF A FORCE: $M = F \times d$, and the PRINCIPLE OF MOMENTS: 'For a body in rotational equilibrium, the sum of clockwise moments about any pivot equals the sum of anticlockwise moments about that same pivot.' Learners copy the formula and principle into journals. The teacher then draws the jembe on the board and asks: 'Using the language of moments, why does the long-handled jembe make digging easier?' Learners write a 3-sentence explanation in journals connecting pivot (blade-soil contact), effort force, perpendicular distance (handle length), and moment size. Groups share; teacher cold-calls four learners. Worked example: Farmer Wanjiku applies 40 N effort on a jembe handle 1.2 m from the pivot — calculate the moment. Repeat for 0.3 m handle. Compare. Learners then solve two additional numerical problems independently.	Begin the class data board share by saying: 'I need a brave volunteer scribe.' Accept the first volunteer. After the class data is recorded, pause and say: 'Look at every row on this board. What is the same every time? Discuss with your partner for 60 seconds.' Apply 60-second pair-talk, then cold-call THREE pairs. After establishing the F×d pattern, write on the board: 'MOMENT OF A FORCE, $M = F \times d$. Units: Newton-metres (N·m).' Say: 'This product — force times distance — is what scientists call the moment. It is the turning effect of the force.' Draw the jembe diagram and mark the pivot, effort arrow, and handle length. Ask: 'Translate what we just found into jembe language — the pivot is…? The force is…? The distance is…?' Apply 12-second WAIT TIME. Cold-call four learners (including at least one who gave only a qualitative prediction in Phase 1). For the worked example, say: 'Work this silently for 2 minutes — no talking.' Then cold-call one learner to walk the class through the solution step by step. For the two independent problems, circulate and check working — correct unit errors immediately.	Claim-Evidence-Reasoning (CER) Writing — learners construct a written scientific explanation that explicitly names the claim (longer handle → larger moment), cites evidence (their own F×d data), and provides reasoning (Principle of Moments). This NGSS practice (Constructing Explanations) anchors the abstract formula to the concrete jembe phenomenon and prevents rote memorisation disconnected from meaning.	Teacher collects four randomly selected journals and checks the 3-sentence jembe explanation for: (1) correct use of the word 'moment', (2) identification of the correct pivot point (blade-soil contact), (3) a numerical comparison (larger d → larger M for same F). Learners who omit any of the three elements receive written feedback at the start of Lesson 3. Teacher also checks the two independent numerical problems for correct unit conversion (cm → m) and correct formula application.
Driving	Each learner receives two sticky	Distribute sticky notes or direct	Driving Question Board — Two-	Teacher photographs the DQB (or

Question Board (DQB) Creation 📹 **VIDEO:** [Equation with the variable in the denominator] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Moment of Inertia (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	notes (or writes in a designated 'DQB Journal' section if physical sticky notes are unavailable). On the FIRST note (green / 'I now know…'), learners write one thing they can now explain about the jembe phenomenon using today's evidence. On the SECOND note (orange / 'I still wonder…'), learners write one question today's experiment did NOT answer — for example: 'What if two forces act on the SAME side?' or 'Does the angle at which you push matter?' The teacher reads several notes aloud (anonymised) and clusters them on the class DQB under two columns: EXPLAINED BY MOMENTS and STILL NEED TO INVESTIGATE. The teacher explicitly connects the new 'I still wonder…' questions to the upcoming lessons: 'Your questions about two forces on the same side, and about parallel forces, are exactly what Lessons 3 and 4 will investigate — you are thinking like physicists.'	learners to the DQB journal section. Allow 4 minutes of silent individual writing. Collect notes and read 6–8 aloud (mix of 'I now know' and 'I still wonder'). As you read each 'I still wonder' note, ask the class: 'Hands up if you share this question.' Count hands visibly and say: 'Eight of you share this question — that means it is important and we will investigate it.' Physically place or draw the note in the correct DQB column. Before closing the DQB activity, point to the original driving question posted in Lesson 1 and say: 'Read our driving question again. Has today's evidence moved us closer to answering it? Show me — thumbs up for YES, sideways for PARTIALLY, down for NOT YET.' Count and record the thumb distribution on the board — this is a public metacognitive marker for the class.	Column Clustering: separating 'explained' from 'still investigating' questions makes the learning progression visible and motivates future lessons. The public thumb-vote on progress toward the driving question develops metacognitive awareness of where the class stands in its evidence-gathering sequence and reinforces that science is a cumulative, iterative process.	notes the distribution in a register) to track which questions recur across the lesson sequence. The thumb-vote distribution is recorded as a class-level metacognitive indicator — if more than 40% vote 'NOT YET', the teacher plans an additional bridging activity at the start of Lesson 3 to consolidate today's core learning before advancing.
Model Building Phase 📹 **VIDEO:** [Equation with the variable in the denominator] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Moment of Inertia (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open their Lesson 1 jembe diagram (drawn in journals during the first lesson) — a sketch of the farmer, the jembe, and their initial arrows showing force and effort. Using a RED pen or a contrasting colour, learners add and label the following to their model: (1) the PIVOT point (labelled 'P — blade contacts soil here'), (2) a double-headed arrow along the handle labelled 'd = perpendicular distance (m)', (3) the effort force arrow labelled 'F (N)', (4) a curved arrow around the pivot representing the MOMENT, labelled 'M = F × d (N·m)', and (5) a written sentence at the bottom: 'A longer handle increases d, so	Say: 'Take out the jembe diagram you drew in Lesson 1. It was your FIRST MODEL — your best guess before any data. Now you have evidence. A scientist always revises their model when they get new evidence. Use a different colour to show what you now know.' Allow 6 minutes of individual model revision. Circulate and prompt learners who draw the pivot in the wrong location: 'Where does the blade touch the soil? That is your pivot — mark it there.' For learners who are ready for extension, ask: 'If Farmer Wanjiku shortens her handle by half, what happens to the moment? Write the maths on your model.' After pair-	Progressive Model Revision (NGSS Science and Engineering Practice: Developing and Using Models) — by physically annotating their original Lesson 1 model in a contrasting colour, learners experience model revision as a scientific act, not an erasure of error. The visible difference between the first and second model makes conceptual growth tangible. The closing preview (wheelbarrow in Kisumu) anchors the next investigation in a new Kenyan context while maintaining continuity with the jembe phenomenon.	Teacher uses a 4-point quick rubric during circulation: 4 = pivot correctly placed + d and F correctly labelled + M = F×d written + explanatory sentence present; 3 = three of four elements; 2 = two of four; 1 = one or none. Teacher records scores in register. Any learner scoring 1 or 2 is paired with a scoring-3-or-4 peer at the start of Lesson 3 for a 5-minute peer model-comparison before new instruction begins.

	for the same effort F, the moment M is larger — the blade turns more forcefully.' Groups briefly compare their revised models with a partner and identify one similarity and one difference. Two volunteers share their models with the class under the visualiser or by holding them up.	comparison, cold-call TWO volunteers to share. Display one strong example and narrate: 'Notice how this model now has a pivot, a labelled distance, a force arrow, AND a moment arrow. This model tells a scientific story — that is what we are building toward.' Close by saying: 'In Lesson 3 we will investigate what happens when MORE THAN ONE force acts on the same beam — which is exactly what happens when a wheelbarrow carries a load of maize at a farm in Kisumu.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) Did every group successfully identify the F×d invariance from their own data — or did some groups simply copy the pattern from the board during the class share? If copying occurred, how will you structure data collection in Lesson 3 to ensure authentic discovery? (2) Were learners able to correctly locate the pivot on the jembe diagram (blade-soil contact), or did most place it at the farmer's hands? Misidentification of the pivot is a persistent misconception — if it appeared today, plan a dedicated 5-minute jembe re-examination at the start of Lesson 3. (3) Review the four journals you collected from Phase 3 — did learners' CER explanations use the word 'moment' correctly, or did they revert to informal language ('force is bigger')? If more than two of the four journals lack the formal term, embed a vocabulary reinforcement warm-up in Lesson 3. (4) Check the DQB 'I still wonder…' notes — do learners' questions anticipate anti-parallel forces and multiple loads? If yes, the class is conceptually ready for Lesson 3. If questions are still at the level of 'why does a longer handle help?', the core Lesson 2 concept needs consolidation before advancing. (5) Were all learners meaningfully engaged during the experiment, or did one learner in each group dominate the measuring while others watched? Consider assigning rotating roles (Measurer, Recorder, Calculator, Reporter) in Lesson 3 to ensure equitable participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners balanced a metre rule on a pencil pivot using masses at measured distances on each side. They recorded weight (F in Newtons) and distance (d in cm/m) for at least five balanced configurations and computed F×d for both sides of the pivot in every trial.
What did I learn?	For every balanced configuration, $F_1 \times d_1 = F_2 \times d_2$. This pattern is the Principle of Moments: a body is in rotational equilibrium when the sum of clockwise moments equals the sum of anticlockwise moments about the pivot. The moment of a force is $M = F \times d$ (units: N·m).
How does this explain the phenomenon?	The Kericho jembe farmer's long handle works because the blade-soil contact point is the pivot. A longer handle increases the perpendicular distance d from the pivot to the line of effort. For the same effort force F, a larger d produces a larger moment $M = F \times d$ — meaning a greater turning effect on the blade — so the soil is broken with less physical effort from the farmer.

LESSON 3 (80 minutes (2 × 40-minute lessons run back-to-back or in a double period)): Formalising the Moment Equation: M = F × d and the Principle of Moments

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To formally define the moment of a force, derive and apply the equation M = F × d, state the principle of moments, and use these tools to explain the anchoring phenomenon of the jembe handle.
Knowledge	Learners determine moments of a force at a point (M = F × d, unit: Newton-metre, Nm); calculate the moments of anti-parallel forces; state and apply the principle of moments (sum of clockwise moments = sum of anti-clockwise moments) as a condition for rotational equilibrium.
Skills	Learners use digital devices and/or print media to research moments of a force; solve structured numerical problems of increasing complexity set in Kenyan contexts; annotate and revise their Lesson 1 predictions; construct and update their cumulative explanatory model of the jembe phenomenon.
Attitudes	Learners appreciate the importance of the turning effect of force in everyday life; display integrity while using digital devices to search for information on moments of a force at a point and moments of anti-parallel forces; respect others' opinions while discussing examples involving turning effect of force in daily life.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	Lessons 1 and 2 established the puzzle (the jembe handle length changes effort) and generated qualitative observations and predictions. Lesson 3 is the formal EXPLAIN phase: learners now receive the scientific language and mathematical tools — M = F × d, the principle of moments, and rotational equilibrium — that transform their intuitive observations into a precise, testable explanation. Every worked example and problem is anchored in the jembe phenomenon or equivalent Kenyan contexts, so learners can immediately test whether the equation accounts for what they saw. By the end of this lesson learners can quantitatively answer WHY the long-handled jembe works better, advancing the driving question from 'we noticed it' to 'we can prove it and predict it'.
Safety Notes	No hazardous materials are used. If a physical balance/metre-rule demonstration is conducted, ensure the pivot stand is stable and weights are secured before release. Learners must not lean on or overload demonstration benches. Digital devices must be used appropriately and returned safely.

B. LESSON OVERVIEW

Lesson 3 is the formal EXPLAIN lesson in the 8-lesson evidence-gathering sequence for Sub-Strand 3.1. Building on the qualitative observations and class predictions from Lessons 1–2, the teacher formally introduces the definition of the moment of a force (M = F × d), its SI unit (Newton-metre, Nm), the principle of moments, and the conditions for rotational equilibrium. Learners use digital devices or print media to research moments of a force, watch animations/simulations, then work through a carefully scaffolded set of numerical problems set exclusively in Kenyan contexts: a market vendor's weighing balance at Gikomba Market (Nairobi), a child on a seesaw in Uhuru Park, and — critically — the jembe-handle puzzle from the anchoring phenomenon. Learners revisit their Lesson 1 written predictions, annotate what was correct or incorrect, and revise their cumulative explanatory models. The lesson ends with learners updating the Driving Question Board with new, more precise causal language.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 1 prediction journals (or digital notes). Working individually for 3 minutes, they re-read their original prediction about WHY the long-handled jembe is easier to use and rate their confidence (1 = not sure, 5 = very sure). They then write one sentence beginning: 'I think the key variable that makes the jembe easier is _____ because _____.' Pairs share their sentences with a shoulder partner. The class hears 4–5 responses via cold-call. Teacher records key vocabulary from learner responses on the board under the heading 'Our Words So Far' (e.g. 'distance', 'force', 'leverage', 'grip point'). Learners are told: 'By the end of today you will be able to replace each of these words with precise scientific language AND calculate an exact number that explains the jembe puzzle.'	1. Instruct: 'Open your prediction journal to Lesson 1. Read what you wrote. Silently rate your confidence 1–5 and write one sentence starting: I think the key variable is _____ because _____. You have 3 minutes — go.' Start timer visibly. 2. After 3 minutes call time. Say: 'Turn to your shoulder partner. Each person has 60 seconds to share. Partner 1 goes first — go.' 3. After 2 minutes, cold-call exactly 5 learners using a name-stick or random method: 'Akinyi, what was your sentence?' 'Mwangi, how did your sentence differ from Akinyi's?' Probe at least 2 responses: 'What do you mean by leverage — can you say more?' Allow 10-second WAIT TIME before prompting. 4. As responses come in, write key terms on the board under 'Our Words So Far'. Do NOT correct scientific inaccuracies yet — affirm thinking: 'Great — you are already noticing something important about distance.' 5. Bridge: 'Scientists had the same instinct as you. Today we will learn the exact equation they developed to describe this — and test it on our jembe puzzle.'	Prediction Revisit with Confidence Rating — learners make their prior mental model explicit and personally stake a claim. This activates cognitive dissonance when the formal equation later either confirms or contradicts their language, making the new knowledge memorable. Recording 'Our Words So Far' on the board creates a visible baseline that will be annotated during the Explain phase, showing growth of understanding.	Observable indicator: Every learner has written a prediction sentence. Teacher circulates during the 3-minute individual write and notes (a) whether learners mention distance/position as the key variable (target idea) or focus only on force magnitude (incomplete model). Cold-call responses reveal the range of prior models. Teacher mentally flags learners who say only 'it depends on force' for targeted questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia	Learners work in pairs or trios (depending on device availability). Task A — Digital/Print Research (10 minutes): Using digital devices (ARES platform resources, Kenya Education Cloud, or teacher-provided print extracts), learners search for and record answers to three specific questions written on the board: (1) What is the definition of the moment of a force? (2) What is the formula and unit? (3) What does the principle of	1. Distribute devices or print packs. Write the three research questions on the board. Say: 'You have 10 minutes. Find answers to all three questions using the resources in front of you. Write in your own words — do NOT copy word for word. Go.' Circulate, pause at 3–4 pairs, ask: 'What does the formula actually mean — can you explain it in everyday language?' 2. After 10 minutes: 'Stop research. Open the	Dual-channel evidence gathering (text research + animation sketching) — learners encounter the same concept through verbal/symbolic and visual/spatial channels simultaneously, strengthening encoding. Sketching with labels (rather than passive viewing) forces active processing. Reporting out creates a shared class evidence base that the teacher can then formalise.	Observable indicators: (a) Each pair can write the formula $M = F \times d$ and state its unit from memory after research — teacher checks by scanning books during circulation. (b) Animation sketches correctly show F arrows, d distances measured perpendicularly to the line of action, and labelled M values. Red flag: learners who draw d as a diagonal or who omit the pivot — these learners need targeted re-

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	moments state? Each pair records answers in their exercise books under the heading 'Evidence from Research'. Task B — Animation/Simulation (8 minutes): Learners watch a short animation (teacher-curated or ARES-embedded) showing a seesaw with adjustable loads and pivot positions. They observe that moving a weight further from the pivot increases the turning effect even when the weight (force) stays the same. Learners sketch two frames from the animation — one balanced, one unbalanced — labelling force (F), perpendicular distance (d), and the resulting moment (M). The whole class then briefly shares observations: 3 pairs report out, guided by teacher cold-call.	animation on your device' (or: 'Watch the screen'). Say: 'As you watch, sketch two frames: one where the seesaw is balanced, one where it tips. Label F, d, and M on each sketch.' Play animation twice if needed. Allow 10-second WAIT TIME between first and second viewing for learners to complete labels. 3. Cold-call 3 pairs to share sketches via visualiser or verbal description: 'Fatuma, describe your balanced-seesaw sketch. What values of F and d did you label?' 'Kipchoge, what changed in your unbalanced sketch?' 4. Affirm accurate observations and gently probe inaccuracies: 'You said the force changed — look again at the animation. Did the weight change, or did something else change?' 5. Bridge to Explain phase: 'Your research and sketches have given you the vocabulary. Now I will show you exactly how to use it.'		explanation of perpendicular distance during the Explain phase.
Explain Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher delivers a structured, interactive direct-instruction sequence (25 minutes) broken into four anchored micro-segments: SEGMENT 1 — Formal Definition (5 min): Teacher writes the definition, formula, and unit on the board. Learners copy and annotate in a two-column 'Notes + My Meaning' format. SEGMENT 2 — Worked Example 1: Gikomba Market Balance (7 min): Teacher works through a problem aloud using a think-aloud protocol, narrating every step. A market vendor in Gikomba Market, Nairobi, balances a 4 N bag of dried beans 0.3 m to the left of the pivot of a simple balance arm. What is the clockwise moment of the beans? (M = 4 N × 0.3 m = 1.2	SEGMENT 1: Write on board: 'The moment of a force about a point = Force × perpendicular distance from pivot. M = F × d. Unit: Newton-metre (Nm).' Say: 'Copy this in your two-column format. On the right, write what each part means in your own language. You have 2 minutes.' Cold-call 2 learners to read their 'My Meaning' column aloud. Specifically ask: 'What does perpendicular mean here, Wanjiru?' Wait 10 seconds. If unclear, draw a door diagram on the board showing that pushing near the hinge (small d) vs. near the handle (large d) gives different moments even for same F. SEGMENT 2: Say: 'I will think aloud through this problem. Your job is to listen and write every step	Anchored Worked Example Sequence with Gradual Release — the teacher begins with full modelling (Gikomba), moves to guided practice (Uhuru Park), then releases to independent application (jembe calculation). Each example is set in a Kenyan context that learners recognise, reducing cognitive load on context and maximising it on the new concept. The final return to the phenomenon closes the explanatory loop: the equation is not abstract — it IS the answer to the puzzle they have been holding since Lesson 1. Prediction Journal Annotation is a metacognitive strategy that makes conceptual change visible and personal.	Observable indicators during circulation: (a) Learners correctly multiply F × d (not F + d or F/d). (b) Learners correctly set up the principle of moments as two sides of an equation before solving for the unknown. (c) Jembe calculation: correct answers are (a) 12 Nm, (b) 60 Nm, (c) 5 times greater. Teacher notes any learner who arrives at wrong answers and provides immediate corrective feedback before moving to the DQB phase. Prediction journal annotations show learners crossing out 'it is easier because of leverage' and replacing with 'it is easier because M = F × d — a larger d gives a larger moment for the same F'.

	Nm). To balance, where must the vendor place a 6 N bag of githeri on the right side? (Using principle of moments: 6 × d = 1.2, d = 0.2 m). Learners solve a mirror question independently. SEGMENT 3 — Worked Example 2: Uhuru Park Seesaw (7 min): A child of weight 300 N sits 1.5 m from the pivot of a seesaw in Uhuru Park. A second child of weight 200 N sits on the opposite side. How far from the pivot must the second child sit for the seesaw to balance? Learners attempt this independently before teacher reveals solution. SEGMENT 4 — Return to the Jembe Phenomenon (6 min): Teacher poses the jembe calculation: A farmer in Kericho applies a downward effort of 80 N to a jembe handle. (a) If the grip is 0.15 m from the blade pivot, what is the moment? (b) If the grip is moved to 0.75 m from the pivot, what is the moment? (c) How many times greater is the turning effect in case (b)? Learners calculate, then compare answers to their Lesson 1 predictions. Teacher asks: 'Did your prediction language match what the equation just told you? Annotate your prediction journal — circle what was right, cross what was wrong, add the correct scientific language.' Learners annotate for 2 minutes.	I write.' Work the Gikomba example step-by-step, narrating: 'First I identify what I know: F = 4 N, d = 0.3 m. The formula is M = F × d, so M = 4 × 0.3 = 1.2 Nm. Now for balance, the principle of moments says clockwise moments must equal anti-clockwise moments, so 6 × d_2 = 1.2, therefore d_2 = 0.2 m.' Pose mirror question: 'Now YOU try: same setup but the beans weigh 5 N at 0.4 m. The githeri bag weighs 8 N. Where does it go? You have 3 minutes.' Cold-call 2 learners for answers; ask a third: 'Otieno, do you agree with that answer? Why?' Wait 10 seconds. SEGMENT 3: Write seesaw problem on board. Say: 'Attempt this independently before I work it. 3 minutes — start.' Circulate, identify one learner with correct working and one with an error. Cold-call the learner with the error first: 'Amina, walk us through your working.' After she finishes, cold-call the correct learner: 'Baraka, where do you agree or disagree with Amina?' Use the contrast to consolidate the principle of moments. SEGMENT 4: Write the jembe problem on the board. Say: 'This is the moment you have been waiting for — we can now calculate exactly what we saw in the video. Solve parts (a), (b), (c). 4 minutes.' After 4 minutes, cold-call 3 different learners for each part. Then say: 'Open your Lesson 1 prediction journal. You have 2 minutes: circle what was correct, put a cross through what was wrong, and write the correct scientific term above it. Go.'		
Driving Question	The class DQB (a large poster or digital shared board with the	1. Direct: 'On your sticky note, write either a new claim you can	Driving Question Board Sorting — making the DQB a living artefact	Observable indicators: (a) 'New claim' sticky notes use the terms

Board (DQB) Creation 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	driving question at the top) is visible at the front of the room. Learners individually write on a sticky note (or type into the ARES DQB tool) ONE of the following: (A) a new claim they can now make confidently, using scientific language from today's lesson, OR (B) a new question that today's lesson has raised for them. Learners post their notes on the board. Teacher reads 5–6 notes aloud, sorting them live into two columns: 'What We Can Now Explain' and 'What We Still Wonder'. The class briefly discusses whether the driving question — 'Why does where you push, pull, or grip something determine how easy or hard it is to make it turn?' — can now be partially answered. Teacher asks: 'Can we now write a partial answer to the driving question using the equation M = F × d? Turn to a partner and draft one sentence.' 3 pairs share. Teacher synthesises: 'The turning effect depends on BOTH the size of the force AND how far it acts from the pivot. A longer jembe handle increases d, which increases M — so the same effort produces a much larger turning effect at the blade. We can now prove this with numbers.'	make (start with: I can now explain that…) OR a new question you have (start with: I now wonder…). You have 2 minutes.' Circulate and encourage learners who are stuck: 'Think about the jembe — what can you explain now that you could not explain in Lesson 1?' 2. Collect notes and read 5–6 aloud. Sort them into the two DQB columns as you read. 3. Say: 'Our driving question asks WHY where you grip something determines how easy it is to turn. With your shoulder partner, write one sentence that partially answers this using M = F × d. You have 90 seconds.' Cold-call 3 pairs. 4. Synthesise by writing the partial answer on the DQB board in a different colour: 'Turning effect (moment) = Force × perpendicular distance from pivot. Gripping further from the pivot increases d, which increases M, making turning easier for the same force.' 5. Preview: 'Next lessons will explore what happens when TWO forces both try to turn something — and how engineers in Kenya use this to design tools that balance perfectly.'	that grows across all 8 lessons gives learners a narrative of their own developing understanding. Separating 'What We Can Now Explain' from 'What We Still Wonder' prevents premature closure and motivates future lessons. Writing a partial answer collaboratively models scientific practice: explanations are built incrementally from evidence.	moment, M = F × d, perpendicular distance, and Nm — indicating uptake of formal vocabulary. (b) Partial answer sentences correctly identify both F and d as co-determinants of turning effect. Red flag: notes that say only 'longer handle = easier' without reference to the equation indicate the learner has not yet connected the formal concept to the phenomenon — teacher provides a brief 1-on-1 prompt before the next phase.
Model Building Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners retrieve their cumulative explanatory model diagram begun in Lesson 1 (a sketch of the jembe showing the farmer, the handle, the blade, and the soil). Working individually (5 minutes), they revise and annotate the model to include: (1) a labelled arrow for the effort force F at the grip point; (2) a double-headed arrow labelled d (perpendicular distance from grip to blade pivot); (3) the calculated	1. Say: 'Take out your cumulative model from Lesson 1. You now have new scientific tools. Spend 5 minutes adding four things to your model.' Write the four annotation requirements on the board. Start the timer. Circulate; prompt: 'Njeri, where is d measured FROM and TO on your diagram? Make sure it is perpendicular.' 'Kamau, have you written a moment value using actual numbers from the jembe	Cumulative Model Revision — returning to the same model diagram across multiple lessons makes conceptual change visible and cumulative, mirroring how scientific models evolve. The star-and-step peer feedback protocol builds metacognitive awareness and collaborative scientific discourse. The 'My model now explains… but still cannot explain…' sentence frames	Observable indicators: (a) Revised models show a correctly placed and labelled d arrow (perpendicular from pivot to line of action of F) — not a diagonal or along the handle. (b) Moment values written on the model are numerically correct and include the unit Nm. (c) Journal sentences name a specific remaining gap (e.g., 'what happens when two forces act on opposite sides')

📖 **HTML:** [Moment of Inertia (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	moment value M = F × d written next to the arrow; (4) a brief written statement of the principle of moments in their own words. Learners then share their revised model with a partner and give each other one 'star' (something done well) and one 'step' (something to improve). Teacher selects 2–3 models to display (via visualiser or holding up) and leads a 3-minute whole-class discussion: 'What has changed in our model since Lesson 1? What does it now explain that it could not explain before?' Learners record one 'model improvement' sentence in their journals.	problem?' 2. After 5 minutes: 'Share with your partner. Give one star and one step. 2 minutes.' 3. Select 2–3 models via visualiser. Ask the class: 'What has this learner done really well in their model?' (cold-call 2 responses, 10-second WAIT TIME each). 'What one thing could make it even stronger?' (cold-call 2 more responses). 4. Wrap up: 'In your journal, write one sentence: My model now explains _____ but it still cannot explain _____. This gap is what our next lessons will fill.' 5. Exit preview: 'In Lesson 4, we will investigate what happens when we have TWO turning forces acting at once — think about how a traditional Kenyan weighing balance stays level. That is your thinking task tonight.'	productive uncertainty, directly linking to the next lesson and sustaining engagement with the driving question.	rather than a vague one — indicating readiness for Lesson 4 content on anti-parallel forces and the principle of moments in multi-force systems.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your notes in your professional journal:

1. PHENOMENON CONNECTION: During the jembe calculation in Phase 3, did learners spontaneously connect the numbers (12 Nm vs. 60 Nm) back to what they observed in the Lesson 1 video — or did you have to prompt them? If prompting was needed, consider adding a more explicit bridging question at the start of Segment 4 in future.

2. PRIOR MODEL GAPS: Which incorrect prior ideas appeared most frequently in the Phase 1 prediction revisit — that the force changed, that the weight of the jembe changed, or that 'leverage' is a separate force? Knowing the most common misconception helps you target it earlier in the sequence next time.

3. NUMERICAL FLUENCY: How many learners struggled with the algebra in the principle-of-moments problems (isolating d in 6 × d = 1.2)? If more than one-third of the class struggled, consider inserting a 5-minute algebraic manipulation warm-up at the start of Lesson 4.

4. DQB QUALITY: Did the sticky-note claims in Phase 4 use formal vocabulary (moment, Nm, perpendicular distance), or did learners default to informal language (easier, harder, longer)? The quality of DQB language is a strong indicator of whether the formal concepts from Phase 3 were internalised or only superficially noted.

5. MODEL REVISION DEPTH: Were learners' revised models genuinely more explanatory than their Lesson 1 models — that is, did they move from 'longer handle = easier' (phenomenological) to 'M = F × d; larger d gives larger M for same F' (causal-mechanistic)? If not, Phase 5 of Lesson 4 may need to begin with a targeted model-comparison activity before introducing new content.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners watched an animation/simulation showing a seesaw with adjustable loads and pivot distances, and used digital devices or print media to research the definition, formula, and unit of the moment of a force.

What did I learn?	The moment of a force $M = F \times d$ (unit: Newton-metre, Nm); the principle of moments (sum of clockwise moments = sum of anti-clockwise moments at equilibrium); the condition for rotational equilibrium; and that the Kericho farmer's long-handled jembe produces a moment 5× greater than the short-handled jembe for the same applied force.
How does this explain the phenomenon?	The jembe handle puzzle: the long-handled jembe is easier to use not because the force applied is larger, but because the perpendicular distance d from the grip to the blade pivot is larger, producing a much greater moment $M = F \times d$ at the blade — making it easier to rotate the blade through hard soil with the same muscular effort.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Anti-Parallel Forces and Couples: When Two Pushes Make a Turn	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate anti-parallel force pairs (couples) using metre rules and force meters, measure the net turning effect of a couple, and establish that the moment of a couple = F × perpendicular distance between the forces — advancing their explanation of why the jembe handle length matters.
Knowledge	Learners will be able to: (a) define a couple as two equal, parallel, opposite forces whose lines of action do not coincide; (b) state that the net moment (torque) of a couple = F × d, where d is the perpendicular distance between the two forces; (c) explain that a couple produces pure rotation with zero net translational force; (d) calculate the resultant turning effect of anti-parallel force pairs.
Skills	Learners will be able to: (i) measure forces using newton meters/spring balances and distances using metre rules; (ii) set up a suspended-beam or metre-rule investigation of anti-parallel forces; (iii) record tabulated data and calculate moments of couples; (iv) identify real-life examples of couples in Kenyan tools and machinery (steering wheels, jembe grip, door handles, tap spanners).
Attitudes	Learners will appreciate that understanding anti-parallel forces empowers Kenyan engineers, farmers, and technicians to design tools that minimise effort and maximise rotational effect, promoting innovation and responsible use of resources.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	In Lesson 3, learners established the Principle of Moments for a single force about a pivot. Lesson 4 extends this to systems where TWO forces act together to create rotation — a couple. This directly addresses why a farmer gripping the jembe with both hands at different positions creates a rotational effect, and why widening the grip (increasing d) reduces the effort needed. By the end of this lesson, learners can quantify the turning power of any two-handed or two-force system, building the conceptual bridge toward Lesson 5 (applications and equilibrium).
Safety Notes	Ensure suspended metre rules are secured with clamps on a stable retort stand — never hand-held at height. Spring balances/newton meters should be zeroed before use. Learners must not pull force meters beyond their maximum scale. If nails or hooks are used on the beam, file sharp edges smooth. Learners with wrist injuries should observe rather than pull force meters.

B. LESSON OVERVIEW

Learners begin by revisiting the jembe puzzle: the farmer uses TWO hands on the handle — what does each hand do, and why does spreading the hands apart make turning easier? Through a teacher-led demonstration using a metre rule suspended horizontally from its centre, two spring balances are attached at equal distances on opposite sides and pulled in opposite directions (parallel but opposite forces). Learners predict, then observe, that the beam rotates even though the NET upward/downward force is zero — something new is happening. Small groups then conduct a structured investigation varying the distance between the two opposing forces while keeping force magnitude constant (and vice versa), recording turning effects and discovering that Moment of Couple = F × d. The lesson closes with learners mapping this to real Kenyan tools — steering wheels on matatus, tap wrenches used by plumbers in Nairobi, and the two-handed jembe grip — updating the class Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners view a still image (or teacher sketch on the board) showing two scenarios: (A) a matatu driver turning a steering wheel by pushing one side only with one finger, and (B) the same driver gripping the wheel with both hands on opposite sides and pushing/pulling simultaneously. Individually (2 min silent think), learners write in their notebooks: 'Which method produces a bigger turning effect on the steering wheel — A or B? What would happen if both forces in B were equal but you moved your hands closer together on the wheel?' Learners then share predictions with a partner (Think-Pair-Share, 2 min). Teacher cold-calls 4 pairs: 'Pair near the window — what did you predict and WHY?' [WAIT 12 seconds after posing question before cold-calling.] Teacher scribes 3–4 contrasting predictions on the board without affirming or correcting — labelling them Prediction 1, 2, 3. Teacher connects to phenomenon: 'Remember our Kericho farmer — she uses BOTH hands on the jembe handle. Today we investigate what happens when two forces act together to make something spin.'	1. Display image/sketch of steering wheel scenarios A and B. 2. Say: 'Write your prediction silently — no talking for 2 minutes. I want YOUR thinking first.' 3. After 2 min: 'Turn to your partner — 90 seconds to compare.' 4. Cold-call exactly 4 pairs using a random name stick. Pose: 'Tell me your prediction AND your reason — one sentence each.' [WAIT 12 sec after naming the pair before they must answer.] 5. Scribe predictions verbatim on the board under 'Our Predictions'. 6. Do NOT resolve predictions yet — say: 'We will return to these at the end and see which prediction the evidence supports.' 7. Bridge to phenomenon: 'Our Kericho farmer grips the jembe with both hands. Both hands exert force. Today we find out what two opposite forces do to a turning effect.'	Think-Pair-Share with public prediction scribing. This strategy externalises prior conceptions so learners have a concrete claim to test against evidence, creating cognitive dissonance when results differ — the engine of conceptual change.	Teacher scans written predictions during pair-share. Observable indicator: Can the learner give a reason (even if incorrect) for their prediction, referencing force and/or distance? Red flag: learners who write 'I don't know' — teacher makes a note to pair these learners with strong explainers during the investigation.
Observe Phase 📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	PART A — Teacher Demonstration (12–22 min): Teacher sets up a metre rule suspended horizontally at its 50 cm mark from a clamp stand (pivot at centre). Two spring balances are attached — one at the 20 cm mark pulling downward, one at the 80 cm mark pulling upward (or vice versa), both with equal force	PART A: 1. Before demonstrating, say: 'I need three volunteers to read the spring balance values aloud as I apply the forces — Achieng, Mwangi, and Fatuma, please come forward.' 2. Apply forces slowly — pause and ask the class: 'What do you SEE happening?' [WAIT 10 sec.] Cold-call 2 learners at the back. 3.	Structured Inquiry with Contrastive Cases. By having half the class vary F and half vary d, learners generate complementary data sets. When groups share in the next phase, the two data sets combine to reveal the multiplicative relationship Moment = F × d — no single group had to do everything, but together they	Teacher checks results tables during circulation. Observable indicators: (1) F × d values are consistent within each row (correct multiplication); (2) learners correctly identify which variable they are changing and which they are controlling; (3) sketch graphs show a linear trend. Red flag: groups where F × d values are

📖 **HTML:** [Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	readings (e.g., 2 N each). Teacher asks: 'Both forces are equal. Net force up/down = zero. Will the rule move?' [Pause 10 sec, take 3 quick hand-votes: yes/no/unsure.] Teacher then applies the forces — the rule ROTATES. Learners record the observation: 'The beam rotated even though the net translational force was zero.' Teacher labels this on the board: 'A COUPLE — two equal, parallel, opposite forces with different lines of action.' Teacher then demonstrates: same 2 N forces but moves them CLOSER together (e.g., 30 cm and 70 cm) — the rotation is less vigorous. Learners record this qualitative observation. PART B — Group Investigation (22–45 min): Groups of 4 receive: 1 metre rule, 1 clamp stand, 2 spring balances (or newton meters), string, and a data recording sheet. Each group investigates ONE of two relationships (half the class each): GROUP TYPE X — Keep force F constant (2 N each balance), vary perpendicular distance d between the two forces (use positions: 10&90, 20&80, 30&70, 40&60 cm from pivot). Measure and record the turning effect (calculated as F × d). GROUP TYPE Y — Keep distance d constant (e.g., always 40 cm between force positions), vary force F (1 N, 2 N, 3 N on each balance simultaneously). Measure and record turning effect. Both group types complete a results table: | Trial | F (N) | d (m) | Moment of Couple = F × d (N·m) | Observations |. Groups plot a quick sketch graph of Moment of Couple vs. their variable.	Record class observation on board in learner language, then restate in scientific language. 4. Demonstrate the closer-hands version and ask: 'What changed? What stayed the same?' [WAIT 12 sec, cold-call 3 learners.] 5. Say: 'This pair of forces has a special name — a COUPLE. Write the definition in your notebook as I write it on the board.' PART B: 1. Distribute materials and say: 'Your group has 18 minutes. Read the instructions on your sheet first — all 4 steps — before touching any equipment. I will time 1 minute of reading.' 2. Circulate: stop at each group at least once. Ask a probing question: 'How did you make sure both spring balances read the same force? What would happen if one was larger than the other — would it still be a couple?' [WAIT 10 sec for response.] 3. At 35 min mark, give a 2-minute warning: 'Finish your last trial and start calculating F × d for each row.' 4. At 40 min, call groups to share ONE finding: 'Group X near the door — read me your largest and smallest F × d value and tell me what changed between them.' [WAIT 10 sec.] Cold-call 2 Group X and 2 Group Y reporters.	constructed the full picture. This mirrors how scientific communities share evidence.	inconsistent — teacher asks: 'Did both balances read the same force? Let us check your zeroing.'
Explain Phase	Groups share their data with the	1. As groups report data, say: 'I	Data Aggregation and Pattern	Written problem solutions collected

📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	class (2 min each, 4 groups report). Teacher records a class summary table on the board combining Group X and Group Y results. Class discusses: 'What pattern do you see?' Learners identify that Moment of Couple increases when either F increases OR d increases, and the relationship is: Moment of Couple (τ) = F × d. Teacher introduces the formal definition: 'A couple is a pair of equal, parallel, antiparallel forces. The moment of a couple = F × perpendicular distance between the two forces. Unlike a single-force moment, a couple produces PURE ROTATION — the object does not translate (move sideways), it only spins.' Learners work in pairs on two structured problems: Problem 1 — 'Otieno, a plumber in Kisumu, uses a pipe wrench. He applies 15 N with each hand, with his hands 0.24 m apart on the wrench handle. Calculate the moment of the couple he applies.' [Answer: 15 × 0.24 = 3.6 N·m] Problem 2 — 'A matatu driver in Nairobi applies a couple of 4.0 N·m to his steering wheel. If each hand exerts 10 N, how far apart are his hands on the wheel?' [Answer: d = 4.0 ÷ 10 = 0.40 m] Teacher cold-calls two pairs per problem to share answers and reasoning. Teacher then returns to the board predictions from Phase 1: 'Let us check — which prediction was closest to what the evidence showed? Why?' Learners identify the winning prediction and explain WHY it was correct using the new formula.	want one person from each group to stand and read ONLY the numbers — your partner will write them on our class table.' Use cold-call for the writer: 'Wanjiru, come write Group Y's data on our board table.' 2. After all data is on the board, pose: 'Look at all four rows. What mathematical relationship do you see between the moment, the force, and the distance?' [WAIT 15 seconds — this is a hard question; do not rush.] Cold-call 3 learners. 3. Affirm the pattern and write: τ = F × d. Say: 'This is the moment of a couple. Say it with me: moment of a couple equals F times d.' 4. For problem-solving pairs: 'You have 4 minutes. Show ALL working. Units matter.' 5. After 4 min, cold-call one pair per problem. Say: 'Walk us through your working, step by step.' If wrong, ask the class: 'Does anyone see a different answer? Raise your hand.' [WAIT 8 sec.] 6. Return to board predictions: 'Prediction 2 said the turning effect would increase when the hands move apart — was Prediction 2 right or wrong? Tell me in one sentence why, using τ = F × d.' Cold-call 2 learners who made Prediction 2.	Recognition followed by Worked Example Transfer. Combining class data sets makes the mathematical relationship emerge from evidence rather than being handed down. Immediately applying the formula to Kenyan real-life contexts (plumber in Kisumu, matatu driver in Nairobi) anchors the abstract equation in familiar experience, supporting transfer of learning.	or photographed. Observable indicators: (1) correct formula stated before substitution; (2) correct units (N·m); (3) answer matches worked solution. Exit check: teacher poses an oral 'rapid fire' to 4 randomly chosen learners: 'If I double the distance between the forces in a couple but keep F the same, what happens to the moment?' [WAIT 10 sec per learner.] Expected answer: 'The moment doubles.' Learners who cannot answer are flagged for re-teaching in Lesson 5 warm-up.
Driving Question	Each learner receives one sticky note (or tears a small piece of	1. Distribute sticky notes. Say: 'Two minutes — write one thing: a	Public Knowledge Display with Storyline Mapping. The DQB	Teacher reads all posted notes after the lesson. Categorise: (A)

Board (DQB) Creation 📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	paper). They write ONE of the following, based on what they now understand better: (a) a NEW question that today's lesson raised for them, (b) a connection between today's couple concept and the jembe phenomenon, or (c) a real Kenyan example of a couple they identified. Learners post their notes on the class DQB in three columns: 'We now understand…', 'We are still wondering…', 'New connections to the Jembe Puzzle'. Teacher reads 4–5 notes aloud and facilitates a 3-minute whole-class discussion: 'Kibii wrote: if a couple produces ZERO net force, why does the jembe blade still go INTO the ground — is that a couple or not?' Class discusses briefly (teacher does not resolve — flags it for Lesson 5). Teacher draws a bold arrow on the DQB from the Lesson 3 card ('Principle of Moments: one force') to the new Lesson 4 card ('Moment of a Couple: two anti-parallel forces') — showing the storyline is building.	new question, a connection to the jembe, or a real Kenyan example of a couple. This is YOUR thinking — no wrong answers.' 2. As learners post notes, group them loosely into the three columns. 3. Read 4–5 notes aloud: 'Amina wrote this connection — listen.' After reading, ask: 'Does anyone want to respond to Amina's idea?' [WAIT 10 sec.] 4. If learners post a note questioning whether the jembe involves a couple, say: 'Excellent — that is exactly the question Lesson 5 will investigate. I am leaving that question right here on the board.' [Physically circle it.] 5. Draw the story-line arrow and say: 'Each lesson, our model of why the jembe handle matters gets more complete. Today we added couples. What do you think is still missing from our full explanation?'	makes learners' evolving understanding visible to the whole class and to themselves. The teacher's explicit arrow-drawing demonstrates epistemic progression — showing that science builds cumulatively — and models metacognition about the learning journey.	correct conceptual connection to couple — indicates understanding; (B) surface or procedural connection (e.g., 'I learned F × d') — indicates shallow understanding, needs extension; (C) persistent misconception visible in note (e.g., 'a couple has a bigger net force') — flag for targeted re-teaching. Aim: ≥70% of notes in category A or B.
Model Building Phase 📹 **VIDEO:** Video: Balancing Act Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12	Learners open their cumulative scientific model (started in Lesson 1 — a diagram of the jembe and farmer). Working individually for 5 minutes, they ADD to their existing model: (1) a new annotated diagram showing the farmer's two hands on the jembe handle as a COUPLE — labelling Force F (left hand), Force F (right hand, opposite direction), perpendicular distance d between the hands, and the calculated moment of the couple $\tau = F \times d$; (2) a written caption: 'When the farmer grips the jembe with both hands spread wide apart, the couple moment = F × d. Increasing d (wider grip) increases the moment, making it	1. Say: 'Take out your jembe model from Lesson 1. Today you add Layer 4: the couple.' Display a reference annotated diagram on the board (drawn quickly by teacher) showing the two forces and distance d on a handle. 2. Say: 'You have 5 minutes. Your diagram MUST show: two force arrows in opposite directions, the label d between them, and the equation $\tau = F \times d$ with a number substituted.' 3. Circulate rapidly — spend 20 seconds per learner, check for: opposing arrows, d label, equation. Stamp correct models. 4. For early finishers, pose the halving-the-grip extension verbally. 5. At 79 min,	Cumulative Model Revision (Layered Modelling). Each lesson adds an explanatory layer to the same physical model — making conceptual growth visible and tangible. Adding the couple annotation forces learners to integrate today's learning with prior knowledge rather than storing it as an isolated fact.	Physical model inspection during circulation. Observable indicators: (1) two force arrows drawn in opposite, parallel directions on the handle; (2) distance d correctly labelled between the force arrows (not from a pivot); (3) $\tau = F \times d$ written with at least one numerical substitution. Learners missing all three indicators are asked: 'Show me where the two forces are — point to them on your diagram.' If learner cannot locate them, teacher re-draws the reference diagram with the learner one-on-one in the last 90 seconds.

🔍 Search ARES for similar readings	easier to rotate the blade into the soil without changing how hard each hand pushes.' Teacher circulates and stamps/initials models that correctly show the couple annotation. Learners who finish early are challenged: 'On your model, show what happens to the couple moment if the farmer HALVES the distance between her hands — rewrite the caption for that scenario.'	close: 'Hold up your model. You should now have four layers: (1) single force moment, (2) Principle of Moments, (3) lever classification, (4) couple moment. Each layer explains MORE of why the Kericho farmer's jembe handle length matters. Next lesson we bring it all together to explain the FULL phenomenon.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) DATA SHARING — Did groups generate consistent enough data for the class table to clearly show $\tau = F \times d$? If values were scattered, consider pre-measuring and marking force positions on the metre rules with tape for Lesson 5 revision. (2) MISCONCEPTION CHECK — Did any learners confuse 'distance between forces' with 'distance from a pivot'? This is the most common error in couple calculations. If more than 3 learners made this error in the model-building phase, open Lesson 5 with a side-by-side comparison of the two distance types. (3) PHENOMENON CONNECTION — When learners updated the DQB, how many correctly identified the two-handed jembe grip as a couple versus a single-force moment? If fewer than half connected it, the bridge between the investigation and the phenomenon may need a more explicit teacher-facilitated link at the start of Lesson 5. (4) KENYAN CONTEXT RICHNESS — Were learners able to generate their own Kenyan couple examples (tap wrenches, steering wheels, bicycle pedals on a crank, sufuria lid handles)? If examples remained confined to the teacher's prompts, plan a 'spot the couple' walk around the school compound at the start of Lesson 5. (5) TIMING — The investigation (Part B) is the most time-variable element. If groups needed more than 18 minutes, consider pre-setting one apparatus per group before learners arrive.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a suspended metre rule rotate when two equal, opposite (anti-parallel) forces were applied at different positions along it — even though the net translational force was zero. They observed that moving the two forces further apart caused greater rotation, and reducing the separation reduced the rotation.
What did I learn?	Learners learned that a COUPLE is a pair of equal, parallel, antiparallel forces whose lines of action do not coincide. The moment of a couple = $F \times d$ (where d is the perpendicular distance between the two forces). A couple produces pure rotation with zero net translational force. Real-life Kenyan examples include a matatu steering wheel, a pipe wrench used by a Kisumu plumber, and the two-handed jembe grip.
How does this explain the phenomenon?	The Kericho farmer's two-handed jembe grip forms a couple. By spreading her hands further apart (increasing d), she increases the moment of the couple ($\tau = F \times d$) without pushing harder (F unchanged). This greater rotational effect drives the blade into the soil more powerfully. This explains why widening the grip — or lengthening the handle — reduces the perceived effort needed to turn the jembe.

LESSON 5 (40 minutes): Calculating Moments of Anti-Parallel Forces — Numerical Problems	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate understanding of anti-parallel forces by calculating their moments in structured numerical problems drawn from Kenyan everyday contexts.
Knowledge	Learners will be able to: (a) determine moments of a force at a point; (b) calculate the moments of anti-parallel forces; (c) apply the turning effect of force in real-life situations.
Skills	Critical thinking and problem solving: as the learner solves numerical problems involving both moments of a force and moments of anti-parallel forces and identifies areas where moments of a force is used in day-to-day life.
Attitudes	Integrity: the learner displays honesty while using digital devices to search for information on moments of a force at a point and moments of anti-parallel forces. Respect: the learner respects others' opinions while discussing with peers on the examples involving turning effect of force in daily life.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	In Lesson 4 learners gathered evidence that anti-parallel forces can create opposing moments on a beam or tool. In this lesson they move from observation to precise quantitative reasoning: they use the moment formula ($M = F \times d$) to calculate clockwise and anticlockwise moments for paired forces acting on objects they already know — a jembe handle, a matatu door handle, an ox-cart axle, and a steering wheel. This advances the driving question by showing that 'how easy it is to make something turn' is not just felt; it can be calculated, predicted, and engineered.
Safety Notes	No hazardous materials in this lesson. Learners work with pencil, paper, and calculators only. Ensure desks are clear of bags to allow pair-working space. If a metre rule is used for demonstration, warn learners not to swing it.

B. LESSON OVERVIEW

Learners open by revisiting the jembe phenomenon — the teacher poses a specific numerical version: 'If the farmer grips the jembe 0.8 m from the pivot and pushes with 50 N, while friction at the blade applies 20 N just 0.1 m from the pivot on the other side, will the jembe rotate?' This hooks the lesson into familiar ground. Learners then work through three progressively complex problem sets (Kenyan contexts: ox-cart axle in the Rift Valley, matatu door handle, steering wheel) in pairs using the Predict–Calculate–Check cycle, with teacher cold-calling a minimum of 4 learners per problem set and enforcing a 30-second wait time. A whole-class sensemaking discussion explicitly targets the two most common misconceptions: (1) forgetting to assign direction (clockwise vs. anticlockwise) to each moment, and (2) confusing the perpendicular distance with the length of the force arrow. The lesson closes with learners updating the class Driving Question Board and revising their personal explanatory models.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners read the 'Jembe Number Puzzle' projected on screen or	Display the jembe diagram with labelled forces and distances. SAY	Predict–Commit–Revisit (PCR): By committing a written prediction	Circulate during the 2-minute silent period. Look for learners who: (a)

📹 **VIDEO:** [Equation with the variable in the denominator] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	written on the board: 'Kamau grips a jembe 0.80 m from the blade pivot and pushes DOWN with 50 N. The soil resists the blade with an upward force of 20 N acting only 0.10 m from the same pivot. Will the jembe blade dig into the soil — and by how much more turning effort does Kamau have compared to the soil resistance?' Learners write individual predictions in their exercise books: (a) which direction they think the jembe will rotate, (b) a rough estimate of the two moments, and (c) what they think will happen to their answer if Kamau moves his grip 0.20 m closer to the pivot. No calculation is expected yet — learners commit to a reasoned qualitative guess.	EXACTLY: 'Before we calculate anything, I want you to THINK first. Look at the diagram. In your book, write: the direction you think the jembe will rotate, a rough guess at how big each turning effect is, and what you think will change if the grip moves closer to the blade. You have 2 minutes — silent, individual thinking.' Start a visible 2-minute timer. After time, cold-call 2 learners (not volunteers — use the class register, rows 1 and 3): 'Aisha, tell me your prediction for direction of rotation.' WAIT 30 seconds before prompting. 'Brian, give me your rough estimate — which moment do you think is bigger?' WAIT 30 seconds. Do NOT correct answers yet — record both predictions on the board under the heading PREDICTIONS so the class can revisit them after calculation.	before instruction, learners activate prior knowledge from Lessons 1–4 about moment direction and magnitude. Recording predictions publicly on the board creates cognitive dissonance when calculations later confirm or refute them, which research shows strengthens retention of the correct concept.	correctly identify the anticlockwise moment as larger (indicating sound prior understanding), (b) assign no direction to their estimates (flag for targeted questioning in Phase 3), or (c) confuse 'force is bigger so moment is bigger' without accounting for distance (key misconception to address). Note 2–3 names for cold-calling in Phase 3.
Observe Phase 📹 **VIDEO:** [Equation with the variable in the denominator] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs through three graded problem sets, each set on a separate card or projected slide. All problems are drawn from Kenyan contexts and share the same structure: a diagram, given forces and distances, and three sub-questions — (i) calculate each moment with direction, (ii) find the net moment, (iii) state what physically happens to the object. PROBLEM SET A (Entry level — Jembe): Same numbers as the Predict phase so learners can immediately check their gut feeling. PROBLEM SET B (Intermediate — Matatu door): 'A conductor pushes a matatu sliding door FORWARD with 35 N at a point 0.60 m from the hinge, while a stiff spring pulls it BACK with 15 N at a point 0.40 m from the same hinge. Calculate the net moment	Distribute or project Problem Set A. SAY: 'Work in pairs. Both of you must write the full solution in your own book — no copying, peer-checking only. Use this structure for every problem: Step 1 — draw and label the pivot. Step 2 — calculate each moment with M = F × d. Step 3 — label each moment CW or ACW. Step 4 — find the net moment. Step 5 — state what happens physically.' Allow 4 minutes for Set A. COLD-CALL 4 learners (use row 4 and row 6 of register) to present Set A answers: 'Zawadi, read us Step 3 only — what direction is Kamau's moment?' WAIT 30 seconds. 'Otieno, do you agree with Zawadi? Why or why not?' WAIT 30 seconds. After Set A consensus, move to Set B (4 minutes) then Set C (5 minutes).	Worked-Example Fading with Pair Accountability: Set A is closest to the demonstrated phenomenon (reducing cognitive load), Set B introduces a new Kenyan context (matatu door) to test transfer, and Set C extends to two separate forces on a distributed object (ox-cart). The fading of scaffolding across sets mirrors NGSS Science and Engineering Practice 5 (Using Mathematics and Computational Thinking). Pair accountability (each learner writes independently) prevents passive copying while enabling immediate peer feedback.	During pair work, circulate and look for: (1) learners who write M = F × d but omit direction labels — prompt with 'Which way would this object spin if ONLY this force acted?' (2) learners who use total bar length instead of perpendicular distance to pivot for the ox-cart problem — address whole-class after cold-call. Collect 4 exercise books (one from each cold-called pair) at end of phase to review written working for feedback in next lesson.

	about the hinge. In which direction does the door move?' PROBLEM SET C (Extension — Ox-cart axle in the Rift Valley): 'Two oxen yoked to a cart apply horizontal forces to a crossbar of length 1.20 m. Ox A (left) pulls FORWARD with 400 N at one end. Ox B (right) pulls BACKWARD with 300 N at the other end. The pivot is at the midpoint of the crossbar. Calculate: (i) moment due to Ox A, (ii) moment due to Ox B, (iii) net moment, (iv) direction the cart axle tends to rotate.' Pairs must show all working: M = F × d, direction label, unit (N·m).	For Set C cold-call 4 DIFFERENT learners. Total observe phase: approximately 15 minutes. Write the correct solutions on the board ONLY after cold-calling, not before.		
Explain Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher explicitly addresses the two most common misconceptions identified during Phase 2, using the board solutions as the anchor. Learners participate in a structured 'Error Autopsy': two deliberately wrong worked solutions are shown (one with missing direction, one using the wrong distance) and learners must identify, name, and correct the errors. After the Error Autopsy, learners write a 'Concept Summary Box' in their notebooks: a generalised rule for calculating net moment of anti-parallel forces, expressed in their own words, with a labelled diagram of their choice (jembe, matatu door, or ox-cart). Finally, learners return to the PREDICTIONS recorded in Phase 1 and publicly confirm or correct them using their calculated answers.	Show Error Autopsy Problem 1 (wrong solution: student labelled both moments as 'clockwise' for a steering wheel turned by two hands in opposite directions). SAY: 'Look at this solution. Something is wrong. You have 30 seconds to find the error — silent, individual.' WAIT 30 seconds. Cold-call: 'Fatuma, what is the error?' WAIT 30 seconds. 'Mwangi, can you explain WHY that error leads to a wrong answer?' WAIT 30 seconds. Repeat for Error Autopsy Problem 2 (wrong distance used — slant length of force arrow instead of perpendicular distance to pivot). After both autopsies, SAY: 'In your Concept Summary Box, write the rule in your own words. You have 3 minutes.' Cold-call 2 new learners to read their rules aloud. Then point to the PREDICTIONS on the board: 'Let's go back to what Aisha and Brian predicted. Aisha said [read prediction]. Was she right? Aisha, tell us — using your calculation.' WAIT 30 seconds.	Error Autopsy (a form of Productive Failure analysis): Presenting wrong solutions before confirming correct ones forces learners to activate their understanding to diagnose the mistake, which is more cognitively demanding — and more durable — than passively reading a correct solution. This directly addresses NGSS Practice 6 (Constructing Explanations) and targets the specific sign/direction misconception documented in the lesson storyline. Connecting back to the original jembe phenomenon keeps the explanation grounded in the anchoring context.	The Concept Summary Box is a written formative artefact — circulate and read 6–8 boxes during the 3-minute writing period. Indicators of mastery: learner states BOTH force AND perpendicular distance, learner explicitly mentions clockwise vs anticlockwise labelling, learner's diagram shows pivot clearly. Indicators of lingering confusion: learner defines moment only as 'force times length' without specifying perpendicular distance to pivot, or omits direction entirely. Flag these learners for the teacher's post-lesson reflection notes.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Equation with the variable in the denominator](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	The class DQB (a large poster or section of the board divided into: WHAT WE OBSERVED / WHAT WE NOW KNOW / WHAT WE STILL WONDER) is updated collectively. Each pair contributes one sticky note (or a written contribution on the board) to at least one column. Learners specifically link today's calculations back to the driving question: 'Why does where you grip or push determine how easy it is to make something turn?' The teacher guides the class to notice that today's lesson has added a QUANTITATIVE layer to the answer — we can now not only say 'further away makes it easier' but also 'exactly how much easier, in N·m'. Learners also add at least one new 'WHAT WE STILL WONDER' question, e.g. 'What happens when the forces are NOT perpendicular to the beam?' or 'How do engineers decide the exact handle length for a jembe?'	SAY: 'We have been asking: why does where you grip the jembe change how hard the work feels? We started with a feeling. In Lessons 1 to 4 we named it — moment. Today we can calculate it. Look at our DQB. In pairs, write one thing we NOW KNOW that goes in the middle column — it must include a number or a formula. You have 90 seconds.' After pairs write, cold-call 2 pairs to read their WHAT WE NOW KNOW contribution. Then SAY: 'Now write one thing you STILL WONDER — something today's calculations cannot yet answer.' Cold-call 2 learners to share their wonder questions and physically attach them to the DQB. SAY: 'These wonder questions will guide Lessons 6 and 7 where we look at real applications — bridges, wheelbarrows, and the steering on a Rift Valley bus.'	Three-Column DQB Tracking: Separating observations, established knowledge, and lingering questions prevents learners from treating each lesson as isolated. The requirement that WHAT WE NOW KNOW entries include a number or formula reinforces the shift from qualitative to quantitative understanding that is the core learning goal of Lesson 5. This mirrors NGSS Practice 8 (Obtaining, Evaluating, and Communicating Information) and makes learning progress visible.	Read all WHAT WE NOW KNOW contributions during the 90-second writing period. A mastery indicator is any entry that correctly uses $M = F \times d$ with correct direction language (e.g. 'The net anticlockwise moment of the jembe is 40 N·m − 2 N·m = 38 N·m, so the blade digs in'). A concern indicator is any entry that restates only the formula without connecting it to the phenomenon or a direction. The STILL WONDER questions also serve as diagnostic data: sophisticated questions about non-perpendicular forces or variable pivot points indicate readiness for Lessons 6–7; very basic questions (e.g. 'What is a moment?') signal the need for targeted re-teaching before proceeding.
Model Building Phase 📹 **VIDEO:** [Equation with the variable in the denominator](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX)](#) Source: CK-12 🔍 [Search ARES for similar](#)	Learners return to the explanatory model they began in Lesson 1 (a labelled sketch explaining why the long-handled jembe is easier to use than the short-handled one). They now add a QUANTITATIVE LAYER to the same diagram: (a) label a specific force value and distance on both the long and short handle, (b) calculate and annotate the moment for each, (c) write one sentence connecting the moment calculation to the physical outcome ('The blade rotates because the anticlockwise moment of ___ N·m is greater than the clockwise moment of ___ N·m'). Learners who finish early extend their model to include the matatu door or ox-cart context, showing	SAY: 'Open your model from Lesson 1. It already shows WHY the long handle is easier — today you are going to add the HOW MUCH. Add numbers to your diagram. Make up reasonable values — a 60 N push, a 0.9 m handle. Calculate the moment. Write the sentence at the bottom of your model.' Allow 4 minutes. Circulate and give specific verbal feedback: 'Chebet, your force arrow is correct — now label the perpendicular distance to the pivot, not the slant of the handle.' 'Kipchoge, well done — you have both moments calculated. Now write the one sentence conclusion.' With 1 minute remaining, cold-call 1 learner to hold up their revised	Cumulative Model Revision: Requiring learners to annotate the SAME diagram they started in Lesson 1 makes conceptual growth visible and tangible — learners can literally see how their explanation has become more precise and quantitative over five lessons. This directly enacts NGSS Practice 2 (Developing and Using Models) and creates a personal artefact that doubles as a study tool for summative assessment. Linking the model explicitly forward to Lesson 6 sustains motivation and narrative continuity across the evidence-gathering sequence.	The revised model is the primary summative-formative artefact of the lesson. Collect 6 models at random (keep a record of whose). Assess against three indicators: (1) correct $M = F \times d$ calculation with appropriate Kenyan values — meets expectation; (2) correct direction labels on both moments — meets expectation; (3) a conclusion sentence that explicitly compares the two moments and names the physical outcome — meets expectation. Learners who meet all three exceed the lesson's minimum goal. Use findings to plan targeted support at the start of Lesson 6.

readings	anti-parallel forces explicitly.	model and narrate it to the class in 30 seconds. SAY to the class: 'In Lesson 6 we will use models exactly like this one to analyse real engineering tools used in Kenya — so keep this model safe.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your teacher journal:

1. COLD-CALL QUALITY: Did at least 4 different learners respond per problem set? Were the 30-second wait times observed — or did you feel pressure to fill silence? If you shortened the wait, note which question and plan to return to it.
2. MISCONCEPTION EVIDENCE: During the Error Autopsy, how many learners initially agreed with the wrong direction label before the correction? If more than one-third of the class agreed with the error, plan a brief 5-minute re-entry activity at the start of Lesson 6 using a simple clockwise/anticlockwise sorting card game.
3. KENYAN CONTEXT RELEVANCE: Did learners respond with greater engagement to the matatu door, the ox-cart, or the jembe problem? Note which context sparked the most spontaneous discussion — use that context as the entry point for Lesson 6's application problems.
4. PAIR ACCOUNTABILITY: Did both members of each pair write independently, or did one learner dominate? If copying was observed, restructure pairs for Lesson 6 and consider individual mini-whiteboards for the next problem-solving phase.
5. DQB WONDER QUESTIONS: Review the STILL WONDER contributions. If any learner asked about non-perpendicular forces or variable pivot positions, acknowledge that question publicly at the start of Lesson 6 — it anticipates the content of Lessons 6 and 7 and rewards intellectual curiosity.
6. MODEL REVISION DEPTH: Did learners add genuine quantitative annotations to their models, or did they simply re-copy the formula? If models lacked numbers, begin Lesson 6 with a 3-minute 'model clinic' where exemplary models are displayed (with learner permission) as anchor examples.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners solved three graded numerical problems (jembe, matatu door, Rift Valley ox-cart) using M = F × d, labelling each moment as clockwise or anticlockwise and calculating the net moment to determine the direction of rotation.
What did I learn?	The turning effect (moment) of a force depends on BOTH the size of the force AND its perpendicular distance from the pivot. For anti-parallel forces, the moments act in opposite directions; the net moment determines which way the object rotates. This explains precisely — not just qualitatively — why a longer jembe handle makes digging easier.
How does this explain the phenomenon?	When two anti-parallel forces act on an object, each creates a moment about the pivot: M = F × d. If the anticlockwise moments exceed the clockwise moments, the object rotates anticlockwise — and vice versa. The farmer in Kericho produces a much larger anticlockwise moment with a long handle (large d) than the soil resistance produces as a clockwise moment close to the blade pivot, so the blade digs in with relatively little effort.

LESSON 6 (80 minutes (2 × 40-minute lessons)): Walk-Around and Real-Life Application Mapping: Turning Effect of Force in Our Environment	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To identify, document, and analyse real-life applications of the turning effect of force in the school compound and surrounding environment, and to connect these observations to the anchoring phenomenon of the jembe handle.
Knowledge	Learners will be able to: (a) identify the pivot, effort arm, load arm, and moment arm in real-life turning-effect examples; (b) recall that moment = force × perpendicular distance from pivot; (c) explain how engineers and tool-makers exploit the turning effect of force in Kenyan infrastructure, agriculture, and transport.
Skills	Learners will be able to: (a) observe and sketch at least five real-life turning-effect examples with correctly labelled diagrams; (b) calculate estimated moments for at least two identified examples using measured or estimated distances; (c) present findings to peers using structured group talk; (d) apply the principle of moments to analyse cranes, pliers, and bicycle brakes shown in video clips.
Attitudes	Learners will: (a) appreciate that scientific principles underpin everyday Kenyan tools and infrastructure; (b) demonstrate curiosity and systematic observation during the walk-around; (c) respect peers' findings and contributions during gallery and group presentations.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	In Lessons 1–5 learners built the concept of moments from the jembe puzzle, calculated moments mathematically, and balanced anti-parallel forces on a metre rule. Lesson 6 is the evidence-gathering walk-around: learners now go beyond the classroom to find, document, label, and quantify turning-effect examples in their real school environment and community, directly answering the driving question by mapping the science onto objects they use every day. This lesson consolidates prior learning and generates a rich class evidence base that will feed the Driving Question Board update and the cumulative model revision.
Safety Notes	1. Learners must stay within the agreed school-compound boundary during the walk-around; no learner may leave the school gate without explicit teacher permission. 2. Learners should not touch electrical switchgear, tap fittings under pressure, or any sharp metalwork — sketch and photograph only. 3. If a bicycle is used for demonstration, it must be stationary and on its stand. 4. All learners must wear closed shoes for the outdoor activity. 5. Groups must have a designated timekeeper to ensure return within the 25-minute walk-around window.

B. LESSON OVERVIEW

Learners begin with a rapid Predict phase linking prior lessons to real-world objects. They then conduct a structured 25-minute walk-around of the school compound in groups of four, using an observation guide to find, sketch, photograph (where devices are available), and label at least five turning-effect examples — identifying pivot, effort, load, effort arm, and load arm, and estimating moments where possible. Back in class, groups mount a mini gallery display and present their findings in a 2-minute structured talk. The class then watches three short video clips (crane at a Mombasa port construction site, pliers used by a jua-kali artisan in Nairobi's Kamukunji market, and bicycle brakes on a Kisumu boda-boda) and applies the moment equation to each. The lesson closes with a DQB update and cumulative model revision, explicitly reconnecting every finding to the jembe handle puzzle.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [The future of creativity in algebra] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners sit in their home groups (4 per group). Each learner silently reads the prompt on the board: 'Think about objects you passed on your way to school today — gates, taps, door handles, bicycle brakes, scissors. Which of these do you think involve a turning effect of force? On your own, list THREE objects and for each one write: (a) what turns, (b) where the pivot is, and (c) where you apply your effort.' Learners write individually for 3 minutes, then share predictions within the group for 2 minutes. One spokesperson per group reads one prediction aloud to the class. Teacher records all predicted objects on the chalkboard in two columns: 'Things that TURN' and 'Where we PUSH/PULL'. Learners then receive their Walk-Around Observation Guide (one per learner) and are briefed on what to document.	T writes the prompt on the board before learners arrive. T says: 'Before we go outside, I want you to predict — no right or wrong answers yet — just use what you already know from our jembe lessons.' T allows 3 minutes of silent individual writing — T circulates and does NOT correct, only nods encouragingly. After group share, T cold-calls FIVE different spokespeople (one per group) to read one prediction each. T records exactly the words used by learners on the board. T then says: 'Hold onto these predictions — when we come back from the walk-around, we will check each one.' T distributes the Walk-Around Observation Guide and reads aloud the five column headings: Object name / Pivot location / Where effort is applied / Estimated effort arm length / Estimated moment. T says: 'Your job outside is to be scientists — observe carefully, sketch neatly, and write down numbers where you can estimate them. You have EXACTLY 25 minutes. Groups return to this door at my signal.'	Strategy: PREDICT–OBSERVE–EXPLAIN (POE) Activation. By committing predictions to paper before the walk-around, learners activate prior knowledge of moments (from Lessons 1–5) and create cognitive dissonance targets — specific objects they will now actively look for and evaluate. Recording predictions publicly on the board creates accountability and motivates careful observation.	Observable indicator: Each learner's written list contains at least three named objects with an attempt at identifying a pivot and effort point. Teacher scans written lists during circulation — if fewer than two objects are listed, teacher prompts: 'Think about the gate you came through this morning — what part stays fixed when you push it?' Target: 100% of learners have a written prediction before the walk-around begins.
Observe Phase 📹 **VIDEO:** The future of creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and	Groups of four walk a pre-mapped route around the school compound. Each learner completes their own Observation Guide, sketching and labelling each example found. Required entries per example: (1) object name and location, (2) sketch showing the object, (3) arrow labelled 'pivot', (4) arrow labelled 'effort (F)', (5) arrow labelled 'load (W)', (6) double-headed arrow labelled 'effort arm (d)', (7) estimated measurement of effort	T leads the class outdoors and stations them at the starting point. T says: 'Remember — the farmer in Kericho gripped the jembe far from the blade to make turning easier. As you walk, ask yourself: where is the pivot, and where does someone apply effort? The further the effort from the pivot, the ____?' (WAIT 10 seconds — accept learner call-outs: 'bigger the moment / easier the turning'). T assigns each group a starting zone to avoid crowding. T circulates	Strategy: STRUCTURED FIELD OBSERVATION with ANNOTATED SKETCHING. Requiring labelled diagrams (pivot, effort, effort arm, moment arm) forces learners to apply the formal vocabulary of moments to authentic objects, not just textbook diagrams. The 'how is this like the jembe?' prompt sustains the anchoring phenomenon thread throughout the outdoor phase, preventing the walk-around from becoming a casual stroll.	Observable indicator: Each learner's Observation Guide contains at least five sketches, each with pivot, effort, and effort arm labelled. Teacher spot-checks two guides per group during the walk-around. Red flag: if a group has only drawn objects without any labels, T intervenes immediately — 'Show me on your sketch: put a dot where the object rotates. That dot is your pivot. Now draw an arrow where you would push.' Target: all groups return with at

Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	arm in cm or m (using hand-spans or a ruler if available), (8) estimated moment in N·m (using an assumed effort force or observed load). Groups are required to find at least FIVE different objects from these categories: gates/doors, tap handles or water-pump levers, flagpole halyard cleats, scissor hinges on window stays, wheelbarrow or trolley, crowbar used by groundskeeper, boda-boda bicycle brake lever, seesaw or balance beam in the school garden. Groups photograph examples using one shared device if available. Learners connect each finding to the jembe puzzle by answering on their guide: 'How is this object like the jembe — and how is it different?'	between groups throughout the 25 minutes, spending approximately 5 minutes per group zone. At each group T asks ONE probing question using cold-call (name a specific learner): e.g., 'Amina — your group found the school gate. Point to me exactly where the pivot is.' or 'Otieno — estimate in metres how far your hand is from the hinge when you push the gate open. Now what is the moment if you push with 20 newtons?' T carries a stopwatch and gives a 5-minute warning signal, then a return signal at 25 minutes. T says on return: 'Find your seats — you have 5 minutes to choose the TWO best examples from your guide to present to the class.'		least five labelled examples and at least one moment calculation attempt.
Explain Phase 📹 **VIDEO:** The future of creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Back in class, each group has 3 minutes to mount their TWO best sketches on the board or display area (tape provided). Groups then do a 2-minute structured stand-up presentation: Spokesperson 1 names the object and points to pivot, effort, and load. Spokesperson 2 states the estimated moment calculation aloud (e.g., 'The school gate hinge is the pivot. We pushed at 0.9 m from the hinge with about 15 N of force. Moment = 15 × 0.9 = 13.5 N·m.'). After all group presentations (approximately 15 minutes), the class watches three video clips (each 2–3 minutes): Clip A — a construction crane lifting steel beams at a Mombasa port site; Clip B — a jua-kali artisan in Nairobi's Kamukunji market using long-handled pliers to bend metal; Clip C — a boda-boda rider in Kisumu applying	T manages gallery mounting — groups stick sketches at eye level. T times each presentation strictly at 2 minutes using a visible timer. After each group, T asks ONE clarifying question to the presenting group using cold-call, rotating through different group members: e.g., 'Wanjiru — your group says the tap handle pivot is the tap spindle. If you gripped the very tip of the handle instead of near the spindle, what would happen to the moment?' (WAIT 12 seconds.) T then asks a second learner from a DIFFERENT group: 'Kipkoech — do you agree with that reasoning? Why?' After all presentations, T introduces the video clips: 'We are now going to see three Kenyan examples — a crane in Mombasa, pliers in Kamukunji, and a boda-boda brake in Kisumu. As you watch, fill in the table in your book.' T pauses	Strategy: EVIDENCE-TO-EXPLANATION MAPPING with CROSS-CONTEXT TRANSFER. The gallery + video sequence asks learners to transfer the moment equation across three scales: hand-sized tools (pliers), human-scale infrastructure (gate, tap), and machine-scale engineering (crane). Explicitly connecting each new context back to the jembe prevents fragmented knowledge and builds a unified explanatory framework — the Principle of Moments as a universal rule.	Observable indicator: During presentations, all spokespeople correctly identify pivot and state a numerical moment estimate (even if the force value is assumed). During video table-fill, T scans 6 exercise books for the three-row table — all three clips must be represented. Red flag: if learners write 'effort' without a distance, T prompts: 'A moment needs TWO things — what are they?' Exit check: T asks four cold-called learners to state the Principle of Moments in their own words before moving to Phase 4.

	bicycle brake levers. For each clip, learners individually fill in a 3-row table in their exercise books: Object | Pivot | Effort arm length (estimated) | Why a longer arm helps. Learners compare video examples to their walk-around findings and to the original jembe phenomenon.	each clip at 1 minute to allow writing time, then plays to the end. After Clip C, T cold-calls FOUR learners (different genders and groups) to share their table entries. T writes key values on the board: 'Bicycle brake lever: pivot = brake pivot bolt; effort arm ≈ 12 cm; braking force arm ≈ 2 cm. Moment ratio = 6:1 — why does this matter for the boda-boda rider?' T closes the Explain phase by returning to the jembe: 'Every single object we found today — gate, tap, crane, pliers, brake — follows the same rule as the farmer's jembe in Kericho. What is that rule?' (WAIT 15 seconds — accept 3 cold-called responses, then write the Principle of Moments on the board verbatim.)		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [The future of creativity in algebra]() Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos]() 📖 **HTML:** [Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX)]() Source: CK-12 🔍 [Search ARES for similar readings]()	Learners return to the class Driving Question Board. Each group receives two sticky notes: one GREEN ('Something we can now answer') and one ORANGE ('A new question this lesson raised'). Groups write their entries in 3 minutes and one member posts them on the DQB. The class reads the new entries silently for 1 minute. Learners update their personal DQB tracker in their exercise books, drawing a line from today's new evidence (gate = 13.5 N·m; bicycle brake lever ratio 6:1) to the original jembe question. Learners identify which part of the driving question they can now answer more fully: 'Why does where you push determine how easy it is to make something turn?' and 'How do engineers and farmers in Kenya use this to design better tools?'	T distributes sticky notes. T reads the driving question aloud: 'Why does where you push, pull, or grip something determine how easy or hard it is to make it turn — and how do engineers and farmers in Kenya use this to design better tools?' T says: 'On your GREEN note, write ONE thing from today's walk-around or videos that helps answer this question. On your ORANGE note, write ONE new question today's evidence raised for you.' T gives 3 minutes of silent writing. T cold-calls TWO groups to read their GREEN note and TWO groups to read their ORANGE note before posting. T then says: 'Look at the DQB as a whole — which questions from Lessons 1 to 5 can we now fully close?' T draws a tick (✓) on any question that the class agrees is now answered, and circles any ORANGE notes that will drive Lessons 7 and 8.	Strategy: DRIVING QUESTION BOARD — ITERATIVE KNOWLEDGE TRACKING. The DQB makes the growth of understanding visible and cumulative. Colour-coding (green = answered, orange = new question) helps learners distinguish between consolidated knowledge and productive uncertainty, both of which are essential for scientific thinking. Connecting today's evidence explicitly to the driving question reinforces that every lesson phase has purpose.	Observable indicator: Every group posts at least one GREEN and one ORANGE sticky note. GREEN notes must reference a specific object found today (not a generic statement). T reads all notes after class to identify persistent misconceptions (e.g., learners who write 'bigger force = bigger moment' without mentioning distance) — these are flagged for re-teaching at the start of Lesson 7.

Model Building Phase 📹 **VIDEO:** The future of creativity in algebra Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative explanatory models (begun in Lesson 1 and revised in Lessons 3 and 5). They add a new layer to their model: a central diagram showing the jembe, and surrounding it, at least THREE objects found today (e.g., school gate, bicycle brake, Kamukunji pliers), each connected by a labelled arrow reading 'same principle: Moment = F × d'. Learners add a 'Rule Box' at the bottom of their model page: 'Wherever a force is applied at a distance from a pivot, a turning effect (moment) is created. The further the force from the pivot, the greater the moment for the same force.' Learners also annotate one object with a worked moment calculation. Learners then write a one-sentence 'model update statement' at the top of the page: e.g., 'Today I extended my model to show that the jembe principle applies to gates, taps, cranes, pliers, and bicycle brakes across Kenya.'	T says: 'Open your model pages. Your model started with a jembe in Kericho. Today you have shown that the same physics governs a school gate in this compound, a crane in Mombasa harbour, and a boda-boda brake in Kisumu. Add at least three of today's objects to your model — draw arrows connecting them to the jembe and label the arrow with the principle they share.' T circulates and checks that: (1) at least three objects are added, (2) each connection is labelled 'Moment = F × d', (3) a Rule Box is present. T cold-calls ONE learner to read their model update statement aloud. T closes the lesson: 'In our next lesson we will look at anti-parallel forces — where two moments act on the same object in opposite directions, like a seesaw or a loaded wheelbarrow. Will the same rule still hold? That is your question to think about before Lesson 7.'	Strategy: CUMULATIVE EXPLANATORY MODEL REVISION. Adding new real-world contexts to an existing model (rather than starting a new diagram) makes the universality of the Principle of Moments explicit and visual. The 'Rule Box' distils the conceptual core, and the worked calculation grounds the abstraction in numbers. The model serves as both a learning tool and a formative assessment artefact that the teacher can review.	Observable indicator: Teacher circulates and confirms that each learner's model page has (a) at least three new objects added, (b) labelled arrows with 'Moment = F × d', (c) a Rule Box, and (d) at least one numerical calculation. Teacher collects 5 randomly selected models at the end of the lesson to review overnight and return with written feedback at the start of Lesson 7. Red flag: any model that still shows the jembe in isolation (not connected to other objects) indicates the learner has not yet generalised the principle — targeted re-teaching in Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your professional journal:

1. EVIDENCE QUALITY: Did all groups return with at least five labelled examples, or did some groups collect only drawings without labels? If labelling was weak, consider providing a worked example on the Walk-Around Guide itself in future iterations.

2. MOMENT CALCULATIONS: Were learners able to attempt numerical moment estimates outdoors, or did the absence of measuring tools create a barrier? If so, prepare pre-measured string (1 m marked) as a portable measuring aid for the next walk-around lesson.

3. VIDEO COMPREHENSION: Did the three video clips (crane, pliers, bicycle brake) successfully transfer the moment concept, or did the Mombasa crane video introduce confusion about large-scale engineering that exceeded the learners' current model? If confused, use only the Kamukunji pliers and boda-boda brake clips and add the crane in Lesson 7.

4. DQB ENTRIES: Review the ORANGE sticky notes posted today. Do the new questions cluster around anti-parallel forces, equilibrium, or practical applications in farming? This will directly inform the emphasis of Lesson 7 (anti-parallel forces and the principle of moments in balance).

5. GENDER AND PARTICIPATION: During cold-calling across all five phases, were both male and female learners equally represented? If not, adjust cold-call targeting in Lesson 7.

6. PHENOMENON CONNECTION: Did learners spontaneously connect the walk-around objects back to the jembe, or did they treat each object as a new, unrelated

example? If the phenomenon thread weakened, open Lesson 7 by explicitly asking learners to re-read their model update statements from today before introducing new content.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners walked around the school compound and identified real-life examples of turning effect of force — including the school gate (pivot at hinge), water tap handle (pivot at spindle), window stay scissor hinge, flagpole halyard cleat, and a groundskeeper's crowbar. They also observed video clips of a construction crane at Mombasa port, long-handled pliers at Kamukunji market in Nairobi, and a boda-boda bicycle brake lever in Kisumu. In every case, effort was applied at a distance from a fixed pivot, producing a turning effect.
What did I learn?	The turning effect of force (moment) is not unique to the jembe — it is the same principle operating in gates, taps, pliers, cranes, and bicycle brakes across Kenya. Moment = Force × perpendicular distance from pivot. A longer effort arm produces a greater moment for the same applied force, making turning easier. Engineers and tool-makers deliberately design long handles and lever arms to multiply the turning effect and reduce the effort needed by the user.
How does this explain the phenomenon?	The Principle of Moments explains why the farmer in Kericho grips the jembe far from the blade: the greater the perpendicular distance from the pivot (the blade-soil contact point), the greater the moment produced by the same muscular effort. This same principle is exploited by Kenyan engineers designing crane booms, jua-kali artisans choosing long-handled pliers, and boda-boda manufacturers sizing brake levers — all are maximising the moment arm to make turning forces more effective with less effort.

LESSON 7 (40 minutes): DQB Update and Model Building: Unifying the Turning Effect of Force	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all evidence gathered across Lessons 1 to 6 to revise and complete their jembe model and update the Driving Question Board with scientific explanations grounded in the moment equation.
Knowledge	Learners will consolidate knowledge that the moment of a force M = F x d, that the principle of moments governs rotational equilibrium, and that a single underlying equation explains all turning-effect phenomena encountered across the unit — jembe handles, seesaws, gates, taps, cranes, and bicycle brakes.
Skills	Learners will construct fully labelled scientific diagrams, calculate moments in context, compare and critique peer models, and articulate a coherent scientific explanation linking multiple phenomena to one principle.
Attitudes	Learners will demonstrate intellectual honesty by revising early misconceptions on the DQB, show persistence in refining their models, and appreciate how scientific understanding is built incrementally through evidence.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	This lesson is the evidence-consolidation apex of the eight-lesson sequence. Learners have predicted, investigated, formalised, and applied the moment concept across six lessons. Lesson 7 requires them to bring all strands together: revise their Lesson 1 naive model of the jembe into a fully scientific diagram, answer every question on the Driving Question Board, and explicitly connect the jembe phenomenon to the supporting phenomena of doors, seesaws, wheelbarrows, and cranes. It prepares learners for the individual assessment in Lesson 8.
Safety Notes	No hazardous materials are used in this lesson. If metre rules and masses are used for brief demonstrations during model discussion, learners must ensure hanging masses are secured before releasing and the pivot pencil is on a flat stable surface. Learners move between group tables — remind them to move chairs carefully to avoid trips.

B. LESSON OVERVIEW
Learners open by revisiting their Lesson 1 jembe sketches and the Driving Question Board to recall original questions and naive ideas. In the Predict phase they individually predict which of their Lesson 1 questions they can now fully answer and which still feel uncertain — surfacing metacognitive awareness. In the Observe phase groups compare Lesson 1 models side by side with findings from all previous lessons, identifying gaps and errors. In the Explain phase learners construct revised, fully labelled scientific diagrams of the jembe system and use the moment equation to annotate calculated values. The DQB Update phase sees groups systematically post revised answers to every question on the board, colour-coding new understanding against original ideas. The Model Building phase closes the lesson with a whole-class sensemaking gallery and teacher-facilitated discussion unifying all phenomena under the single moment equation, directly answering the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Each learner receives their original Lesson 1 jembe sketch and a	Distribute learner Lesson 1 sketches before the lesson begins.	Metacognitive self-assessment: learners surface their own prior	Teacher scans annotated sketches for patterns: Are learners still

📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	printed copy of the Driving Question Board questions (or views the class DQB). Working individually for 3 minutes, learners annotate their own sketch with a coloured pen: they circle parts they now know are scientifically correct, cross out parts they now know are wrong, and place a question mark beside anything still uncertain. They then write one sentence predicting: Which question on the DQB do you think you can answer most completely right now, and which question still feels hardest to answer? Learners do not share yet — this is a private metacognitive inventory that will be revisited at the end of the lesson.	As learners annotate, circulate silently and observe — do not intervene or confirm. After 3 minutes say: Without telling anyone your answer yet, hold up your sketch. Look at how much you drew in Lesson 1. Just notice it. Now we are going to find out together how much has changed. Use a 10-second silent wait after this prompt before transitioning. Record on the board: Lesson 1 idea versus Lesson 7 evidence — this column heading will anchor the whole lesson.	model against accumulated evidence. The physical act of annotating their own earlier drawing with a different-coloured pen creates a visible record of conceptual change — a key sensemaking tool in model-based learning.	drawing effort and load at the wrong positions? Are pivot points missing or misplaced? This informs which model errors to prioritise during the Explain phase. Teacher notes names of two or three learners whose sketches show persistent misconceptions for targeted cold-calling later.
Observe Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four are formed. Each group spreads all available evidence on their table: the Lesson 2 metre-rule data tables, Lesson 4 couple investigation notes, and Lesson 6 walk-around sketches or photographs. Groups spend 6 minutes doing a structured comparison task using a three-column chart headed What we observed, What we measured, What the evidence means for the jembe. Each group must find at least three links between their walk-around findings from Lesson 6 and the original jembe puzzle. For example: The school gate pivot is the same type of turning system as the jembe blade in soil. Both have a pivot, an effort arm, and a load arm. Increasing the distance from the pivot increased the moment in both cases. Groups record their three links on a large sheet of paper or mini whiteboard.	Pose the observation prompt aloud and write it on the board: What patterns appear across ALL the evidence you have collected since Lesson 1? You have 6 minutes. Start now. Circulate and ask probing questions to groups that list surface features rather than underlying principles: You wrote the gate has a long arm. What does that tell us about moments specifically? Push groups to use the equation $M = F \times d$ in their links rather than only describing what they saw. At the 4-minute mark give a time warning: Two minutes left — make sure your three links are on your chart in equation form where possible.	Cross-lesson evidence synthesis: by physically placing all prior-lesson artefacts on the table simultaneously, learners treat their own data as a coherent evidence base rather than isolated activities. The three-column chart scaffolds the move from raw observation to mechanistic explanation.	Monitor charts for the quality of links. A strong link names the pivot, states $F \times d$ for both sides, and connects to the jembe. A weak link only describes shape or appearance. Teacher uses this to select two contrasting groups — one strong, one weaker — to share at the start of the Explain phase, making the contrast productive.
Explain Phase 📹 **VIDEO:**	Each learner now independently draws a revised jembe diagram	Before learners begin drawing say: Your diagram must tell the full	Representational refinement: the move from a naive sketch (Lesson	Collect three to four revised diagrams from learners identified

Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	(A4 blank paper or exercise book, large diagram). The diagram must include and label: the pivot point (blade tip in soil), the effort force arrow (hands on handle) with direction and estimated magnitude in Newtons, the perpendicular distance from pivot to effort (effort arm in metres), the load force arrow (weight of soil resisting blade) with direction, the perpendicular distance from pivot to load (load arm in metres), the calculated moment of the effort (M = F x d), and the calculated moment of the load (M = F x d). Learners use realistic Kenyan farm values: effort force approximately 80 N, effort arm 1.2 m for long handle versus 0.3 m for short handle, load force 40 N, load arm 0.1 m. They calculate both moments and write a conclusion: For the long-handled jembe, effort moment = 80 x 1.2 = 96 Nm. Load moment = 40 x 0.1 = 4 Nm. The effort moment far exceeds the load moment, so the blade easily rotates into the soil. For the short-handled jembe, effort moment = 80 x 0.3 = 24 Nm — much closer to or less than the load moment, making digging much harder. Learners annotate: This is why the farmer in Kericho finds the long-handled jembe easier.	scientific story of the jembe puzzle. A scientist who has never heard of our phenomenon should be able to look at your diagram and understand exactly why handle length matters. You have 8 minutes. Begin. After 4 minutes, cold-call one learner whose Lesson 1 sketch showed a misconception: [Name], I noticed in your Lesson 1 sketch the effort arrow was pointing downward from the handle tip. Now that you know about moments, where exactly does the effort act and what is the moment arm? Use 30-second wait time before accepting the answer. If the learner is uncertain, prompt: Think about what d means in M = F x d. What does perpendicular distance mean in the jembe context? After diagrams are complete, pairs peer-review each other using three criteria written on the board: 1. Is the pivot correctly placed? 2. Are both moment arms labelled with values? 3. Are both moments calculated and compared? Pairs award a tick, a cross, or a question mark for each criterion and return the diagram with written feedback.	1) to a fully quantitative scientific diagram is the core sensemaking activity of this lesson. Peer review using explicit criteria forces learners to apply the conceptual framework as evaluators, deepening understanding.	in the Predict phase as having persistent misconceptions. Review during the DQB phase to check whether cold-calling and peer review have resolved the error. Use exit question at end of phase: Ask one learner to state in one sentence why the long-handled jembe works better, using the word moment and a numerical comparison.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Groups return to the class Driving Question Board displayed at the front. The DQB has been up since Lesson 1 and contains all original questions, predictions, and wonderings posted by the class. Each group receives three sticky notes of a different colour from Lesson 1 (e.g., green for confirmed answers, orange for revised answers, red for any	When groups begin posting, circulate and challenge any vague answers: You wrote longer handle means more turning. Can you write that using the equation? Make them be precise. During the gallery walk, stand to the side and observe which questions generate the most crowding or rereading — these indicate lingering uncertainty and should be addressed in the	Driving Question Board update is the signature sensemaking structure of the lesson. It externalises conceptual change, makes the progression from naive prediction to evidence-based explanation visible to all learners, and creates a shared record the class can reference in Lesson 8. The gallery walk adds a quiet reflective dimension after the	Quality of sticky-note answers is the primary formative data source. Teacher photographs the updated DQB at the end of the phase for use in Lesson 8 and for teacher reflection. Any DQB question left with only a red unanswered note signals a gap to address in Lesson 8 consolidation.

for similar videos 📖 **HTML:** Moment of Inertia (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	remaining questions). Groups spend 5 minutes assigning notes to every question on the DQB. For each question they must write a one-sentence scientific answer referencing M = F x d or the principle of moments and, where possible, a specific Kenyan example from the lesson sequence. Sample DQB questions from Lesson 1 that groups address: Why does a longer handle make digging easier? — Answer: A longer handle increases the effort arm d, so for the same force F, the moment M = F x d is larger, producing a greater turning effect on the blade. Why does holding the jembe near the blade make your arms tired faster? — Answer: A shorter effort arm means a smaller moment for the same force, so more effort force is needed to produce enough turning effect, causing rapid muscle fatigue. Is the same idea used in other tools? — Answer: Yes. The school gate, tap handle, bicycle brake lever, and crane arm all use a long effort arm to produce a large moment with manageable force. After posting, the class does a 2-minute silent gallery walk past the DQB, reading all notes. Teacher then facilitates a 3-minute whole-class discussion.	whole-class discussion. Open the discussion with: Look at the board. In Lesson 1 we had questions and guesses. Now we have evidence and equations. Has the science changed the question, or just the answer? Wait 20 seconds before taking responses — this is a high-cognitive prompt. Cold-call at least two learners who have been quieter in previous lessons. Conclude the DQB phase by pointing to the driving question posted at the top of the board and saying: We asked why where you push or grip determines how easy or hard it is to make something turn. What is our class answer now, in one sentence using science? Take two responses. Write the agreed class answer beside the driving question.	active posting.	
Model Building Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups now combine individual revised jembe diagrams into a single group consensus model on A3 paper or a mini whiteboard. The group model must show: the jembe system with all labels; a second panel showing one supporting phenomenon chosen by the group from Lessons 4 or 6 (door handle, seesaw, wheelbarrow, crane, pliers, or	During group model building (approximately 5 minutes) circulate and ask each group: Which supporting phenomenon did you choose and why? Push groups that chose the easiest example (seesaw) to articulate the moment calculation explicitly, not just the concept. For groups that chose the wheelbarrow or crane, affirm the complexity and ask: What is the	Cross-phenomena model comparison is the highest-order sensemaking task in the lesson. By placing the jembe diagram alongside a chosen real-world example on the same sheet and writing a connecting statement, learners perform the scientific practice of generalisation — recognising that one equation describes structurally different	Teacher assesses group models against four criteria: correct pivot placement in both panels, correct effort and load arm labelling with values, correct moment calculations, and a connecting statement that references the equation. Teacher identifies one model that is exceptionally clear and one that has a specific error to use as comparison exemplars at

📖 **HTML:** [Moment of Inertia (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	bicycle brake) with the same labels; and a connecting statement at the bottom explaining in two to three sentences how both panels illustrate the same principle. Groups display models on desks for a 3-minute standing gallery walk. Each group leaves one member as a guide at their model to answer questions from peers passing by. After the gallery walk, teacher facilitates a 4-minute whole-class sensemaking discussion using the prompt: Every model on these desks, whether it shows a jembe or a crane or a bicycle brake, contains the same three things. What are they? Expected responses: a pivot, a force, and a perpendicular distance. Teacher writes on the board: Turning effect = Force x perpendicular distance from pivot. M = F x d. This is the one idea that explains all of them. The lesson closes with teacher returning briefly to the anchoring phenomenon: The farmer in Kericho did not need to know the equation. But now you do. And because you do, you can design a better tool, check an engineers drawing, or explain to your family why the jembe handle length is not an accident — it is science.	pivot in this system? How did the engineer use a long effort arm here? During the gallery walk, prompt visiting learners to ask the guide questions rather than just reading: Ask the guide — what would happen to the moment if the effort arm were halved? During the whole-class discussion use cold-calling with 30-second wait time for the central question about the three common elements. After learners give pivot, force, and distance, press further: In the jembe, what is the pivot? What is d? What is F? Then in the school gate from Lesson 6, what is the pivot? This explicit mapping consolidates transfer. End with the closing statement about the Kericho farmer to return the lesson emotionally and contextually to the anchoring phenomenon.	physical situations. The gallery walk with a resident guide adds oral explanation as a consolidation modality.	the start of Lesson 8. Learners who were guides during the gallery walk are informally assessed on their oral explanations — teacher notes fluency and accuracy for the Lesson 8 oral component.

D. TEACHER REFLECTION

After the lesson, consider the following questions: Were learners able to move from descriptive language (longer handle is easier) to mechanistic language (larger moment arm produces larger moment for same force) when updating the DQB? Did the peer review criteria during diagram construction successfully surface and correct persistent misconceptions about pivot placement and moment arm direction? Which supporting phenomena did most groups choose for their consensus model — does this suggest any gaps in understanding of specific contexts that need revisiting in Lesson 8? Did the 30-second wait times before cold-calling produce richer responses than in earlier lessons, suggesting growing learner confidence? Were there learners who remained passive during the gallery walk or DQB update — how will you ensure these learners engage individually in the Lesson 8 assessment? Review the photographed DQB: are there any questions still marked with a red unanswered note that require explicit teaching in Lesson 8 before the assessment?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed their own Lesson 1 jembe sketches placed alongside all evidence collected across six lessons — metre-rule data tables, couple investigation notes, and walk-around photographs — revealing the gap between their initial naive model and a scientifically complete explanation of the turning effect of force.
What did I learn?	Learners learned that a single equation, M = F x d, unifies every turning-effect phenomenon encountered in the unit: the jembe, the school gate, the seesaw, the bicycle brake, and the crane all involve a pivot, a force, and a perpendicular distance, and increasing the distance always produces a larger moment for the same applied force.
How does this explain the phenomenon?	Learners explained why the farmer in Kericho finds the long-handled jembe easier by constructing a fully labelled scientific diagram showing that the effort moment of 96 Nm far exceeds the load moment of 4 Nm, and by connecting this calculation to the updated Driving Question Board answer: where you push or grip something determines the moment arm, and the moment arm determines the turning effect produced by that force.

LESSON 8 (80 minutes (2 × 40-minute lessons run consecutively)): Consolidation, Assessment and Reflection: The Final Explanation of Turning Effect of Force	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and assess learners' full understanding of the turning effect of force — moments of a force and moments of anti-parallel forces — through unseen numerical problems, a practical demonstration, and personal reflection, culminating in a complete and evidence-based explanation of the anchoring phenomenon.
Knowledge	Learners demonstrate that: (a) the moment of a force = Force × perpendicular distance from the pivot; (b) for a body in equilibrium, the sum of clockwise moments equals the sum of anti-clockwise moments (Principle of Moments); (c) anti-parallel forces constitute a couple whose resultant moment = F × d; (d) handle length determines the perpendicular distance and therefore the moment produced, explaining why the long-handled jembe makes digging easier.
Skills	Learners are able to: (i) solve unseen, multi-step numerical problems involving moments and anti-parallel forces set in a Kenyan context (Lake Victoria fisherman using a boat oar as a lever); (ii) design and carry out a practical balancing task using a metre rule and given masses; (iii) explain and justify results verbally using precise scientific language; (iv) revise and annotate their cumulative model from Lesson 1 to reflect all accumulated understanding; (v) write a structured reflective journal entry linking classroom science to home and farm life.
Attitudes	Learners appreciate that scientific understanding grows iteratively — from puzzlement to prediction to evidence to explanation — and that the physics of moments is not abstract but is embedded in everyday Kenyan tools, farm work, and fishing practice. Learners value perseverance, honest self-assessment, and the ability to explain phenomena clearly to others.
Key Inquiry Question	How is the turning effect of force used in our daily life?
Purpose in Storyline	This final lesson closes the storyline arc that began with the Jembe Handle Puzzle. Learners now possess all the conceptual tools — moment formula, Principle of Moments, couples, and anti-parallel forces — needed to give a complete, quantitative explanation of why the long-handled jembe works so much better. They compare their Lesson 1 naïve model with their final revised model, complete the Driving Question Board, and produce individual written and practical evidence of mastery. The lesson ends with learners committing to notice turning-effect phenomena in their own lives — on the farm, at the fishing dock, in the kitchen — bridging school science and lived Kenyan experience.
Safety Notes	Ensure metre rules are placed securely on the pivot (triangular wedge or pencil) so they do not flip and strike learners. Masses (stones, water bottles, or standard slotted masses) must be tied or clamped to prevent falling. Learners should stand clear when releasing masses. No sharp metal edges on improvised pivots. Remind learners to handle apparatus carefully and return it to the teacher after the practical assessment.

B. LESSON OVERVIEW

Lesson 8 is the culminating lesson of Sub-Strand 3.1. It has three interlocking components delivered across the double period: (1) an individual written assessment on unseen numerical problems involving moments and anti-parallel forces, set in the context of a fisherman at Lake Victoria using a boat oar as a lever; (2) an individual oral/practical assessment in which each learner balances a metre rule with given masses and verbally explains the result using the Principle of Moments; and (3) a whole-class Final Explanation sequence in which learners compare their Lesson 1 jembe model with their final model, complete the Driving Question Board, and write a personal reflection journal entry. The lesson is anchored throughout in the original Jembe Handle Puzzle: every assessment item and discussion prompt returns to the

question of why handle length changes how hard digging feels, and what this means for tool design in Kenya and beyond.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners enter to find the original Jembe Handle Puzzle image displayed on the board alongside their Lesson 1 prediction cards (returned by the teacher). Each learner reads their own Lesson 1 prediction silently for 2 minutes. They then write a one-sentence 'Before and After' statement in their notebooks: 'In Lesson 1 I thought… but now I know…'. They share with their seat partner for 90 seconds (think-pair). The teacher cold-calls 4 pairs — one from each side of the room — to share their 'Before and After' statements aloud. The teacher writes three contrasting 'Before' statements and three 'After' statements on the board in two columns, leaving them visible for the entire lesson as a record of the learning journey.	BEFORE CLASS: Return Lesson 1 prediction cards to learners (pre-sorted). Display the jembe image and the driving question prominently. AT START: Say — 'You wrote these predictions eight lessons ago. Read yours quietly. Do not change anything. Just read.' Allow 2 minutes of silent reading. Then: 'In your notebook, write one sentence: In Lesson 1 I thought… but now I know…. You have 90 seconds.' Set a visible timer. After 90 seconds: 'Turn to your partner. Share your sentence. You each have 45 seconds.' After pair-share, COLD-CALL 4 pairs by name (select deliberately: one high-confidence, one who struggled early, two mid-range). For each, ask: 'What changed in your thinking, and what lesson or activity changed it?' WAIT 12 seconds after each question before accepting answers. Record responses verbatim in two columns on the board: LESSON 1 THINKING | FINAL THINKING. Say: 'Keep your eyes on these columns. By the end of today, every one of those right-hand ideas will be fully explained and tested.'	Reflective Prediction Revisit — learners compare their initial intuitive model with their current evidence-based understanding. This metacognitive strategy makes the learning journey visible and explicit, motivating learners to see that science progresses through iterative revision of ideas rather than sudden revelation. It also activates all prior schema built across Lessons 1–7, priming working memory for the assessment tasks ahead.	Teacher scans the 'Before and After' notebook entries as learners write (circulate for 60 seconds). Observable indicator of readiness: learner's 'After' sentence correctly references moment = F × d or the Principle of Moments without prompting. If more than one-third of the class cannot produce a correct 'After' sentence, spend 3 additional minutes on a rapid whole-class recap (write the key equations on the board) before proceeding to the written assessment.
Observe Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan	Learners complete an individual, closed-book written assessment (25 minutes). The paper contains four items set in Kenyan contexts: ITEM 1 (6 marks) — Ochieng, a fisherman at Lake Victoria, uses a	Distribute the assessment paper face-down. Say: 'This is individual, silent work. You have 25 minutes. Read each question fully before writing. Show all working — answers without working earn zero	Transfer Task Assessment — unseen problems in novel Kenyan contexts (oar, couple, farmer) require learners to transfer, not merely recall, conceptual understanding of moments. The	While circulating, teacher notes: (a) Are learners correctly identifying pivot, effort, and load in Item 1? (b) Are learners showing the Principle of Moments equation in Item 2? (c) Are learners

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	2.4 m wooden oar as a lever to push his boat away from a dock. He applies a force of 80 N at one end; the water acts as a pivot at a point 0.4 m from the blade end. (a) Identify the effort, load, and pivot. (b) Calculate the moment of Ochieng's effort about the pivot. (c) If the water resistance is 480 N, is the oar in equilibrium? Show your working. ITEM 2 (5 marks) — A uniform metre rule is balanced on a pivot at its 50 cm mark. A 3 N weight is hung at the 20 cm mark. Where must a 4 N weight be placed to maintain balance? Show all working and state the Principle of Moments. ITEM 3 (5 marks) — Two fishermen push a beached boat at Kisumu using anti-parallel forces: one pushes with 200 N and the other with 200 N, both perpendicular to the length of the boat, 1.8 m apart. (a) What is the resultant force? (b) Calculate the turning moment of the couple. (c) Will the boat translate, rotate, or both? Explain. ITEM 4 (4 marks) — A farmer in Kericho switches from a jembe with a 0.4 m handle to one with a 0.8 m handle, applying the same 50 N effort. (a) Calculate the moment in each case. (b) Explain in one scientific sentence why the longer handle makes digging easier. Learners work in silence. The teacher circulates, does NOT answer content questions, but may clarify wording.	marks for calculation items. You may use a ruler and calculator.' Flip papers over together on your signal. Set a visible 25-minute timer. CIRCULATE continuously — do not stand at the front. Walk every row every 5 minutes. If a learner is stuck on Item 1 for more than 8 minutes, quietly place a prompt card on their desk that reads: 'Moment = Force × perpendicular distance from pivot.' (This is a scaffold, not an answer.) At 20 minutes, give a '5 minutes remaining' warning. At 25 minutes: 'Pens down. Turn your paper face-down.' Collect all papers promptly. Do NOT discuss answers immediately — say: 'We will see how well your understanding holds up in the practical next, then we will revisit any questions during reflection.'	deliberate use of Lake Victoria and Kericho contexts signals that the physics is not confined to a classroom but operates in real Kenyan work and life — directly connecting to the driving question.	recognising zero resultant force but non-zero moment in Item 3? These observations inform the teacher's targeted feedback during the Final Explanation phase. Papers will be formally marked after class; however, teacher's circulating observations provide immediate formative data to shape the closing discussion.
Explain Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English	PART A — PRACTICAL ASSESSMENT (15 minutes, staggered): While most of the class completes a structured peer-review task (see below), the teacher calls learners one at a time to the demonstration bench.	PRACTICAL SETUP: Before the lesson, prepare 8 practical stations (metre rule + pencil + masses) so that multiple learners can be assessed in rotation if the class is large (adjust timing for class size; for 40 learners, run two	Two-stage Explanation Construction — the practical assessment first requires individual application of the Principle of Moments in a physical, manipulable context, consolidating procedural and conceptual	PRACTICAL: Observable indicators of mastery — learner places mass at correct position (±2 cm), states 'sum of clockwise moments = sum of anti-clockwise moments' unprompted, and connects moment formula to

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner is given a metre rule balanced on a pencil pivot at the 50 cm mark, two known masses (e.g., a 100 g stone tied with string at a known position), and a third unknown mass. The learner must: (i) hang the two known masses at specified positions, (ii) find the position to hang the third mass to achieve balance, and (iii) verbally explain what they did using the Principle of Moments ('The sum of clockwise moments equals the sum of anti-clockwise moments'). Teacher awards marks for correct placement AND for verbal explanation. While waiting, learners complete a peer-review of a model answer to Item 2 (printed on a card), marking it using a provided 3-point rubric. PART B — WHOLE-CLASS FINAL EXPLANATION (10 minutes): Teacher reconvenes the full class. Displays the anchoring phenomenon image again. Says: 'You now have all the science. Give me the complete explanation of the jembe puzzle — in three sentences: what moment is, why handle length matters, and what this means for tool design.' COLD-CALLS 3 learners. After each, asks the class: 'What would you add or change?' Builds a consensus 'Final Explanation' statement on the board, written collaboratively, using precise scientific language. This statement is copied by all learners into their notebooks as the class's definitive answer.	simultaneous assessment stations). At each station, ask exactly these three questions: (1) 'Place the third mass so the rule balances. Tell me what you are doing as you go.' (2) 'Why does it balance at that position? Use the principle by name.' (3) 'How does what you just did explain the jembe puzzle?' WAIT 10 seconds after each question. Award full marks only if learner uses 'Principle of Moments' by name and correctly relates distance to moment. WHOLE-CLASS: Write the driving question on the board in large text. Cold-call Learner A: 'Give me sentence one — what is a moment?' WAIT 12 seconds. Cold-call Learner B: 'Sentence two — why does handle length matter in the jembe?' WAIT 12 seconds. Cold-call Learner C: 'Sentence three — what does this mean for how engineers or farmers should design tools?' WAIT 12 seconds. After each, ask the class: 'Thumbs up if you agree, thumbs sideways if you'd change something.' Refine until the class consensus statement is scientifically accurate. Write the final 3-sentence explanation on the board. Read it aloud together. Say: 'This is the answer to the question that puzzled you in Lesson 1. You built this understanding yourselves — from a jembe in a school garden to the physics of every lever, oar, and door handle in Kenya.'	understanding. The whole-class Final Explanation then elevates individual understanding to a shared, socially negotiated scientific explanation — mirroring how science actually works. The explicit three-sentence structure (definition → mechanism → application) scaffolds scientific communication without removing the cognitive challenge.	jembe handle. Indicator of partial understanding — correct placement but only restates the formula without explaining why distance matters. Indicator of insufficient understanding — cannot balance the rule or cannot explain beyond 'because it is heavier.' Teacher records outcomes in a quick checklist. WHOLE-CLASS: If the consensus Final Explanation statement required more than two rounds of revision, teacher flags this as a gap for individual follow-up.
Driving Question Board (DQB) Creation 📹 **VIDEO:**	The Driving Question Board (maintained since Lesson 1) is brought to the front of the class. Learners review all sticky notes added across Lessons 1–7 —	Retrieve the DQB from its display location. Lay it flat or prop it where all learners can see. Say: 'Look at every question your class asked since Lesson 1. Your job in the	DQB Closure Protocol — systematically answering, connecting, and extending questions on the DQB makes the growth of understanding visible to	Teacher checks: (a) Are all major conceptual questions on the DQB now answered correctly? (b) Are the 'Connections to Real Life' notes specific and scientifically

Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	questions, partial answers, and connections to the phenomenon. Working in their home groups (same groups from Lesson 1), learners spend 5 minutes doing three things: (a) move any remaining unanswered question notes to a 'Now Answered' column and write the answer on the back; (b) add one new note to a 'Connections to Real Life' column naming a specific place, tool, or activity in their own community where the turning effect of force operates; (c) add one 'Remaining Wonder' note — a question about moments or rotational physics that this sub-strand did not fully address and that they would like to explore further. Groups share their 'Connections to Real Life' notes aloud (30 seconds per group, teacher calls on all groups). The board is then photographed (by teacher) and the photograph is saved as a class artefact. The teacher reads aloud the very first driving question posted in Lesson 1 and asks the class to answer it together in one sentence.	next 5 minutes — in your original groups — is: first, answer every question that is still unanswered and move it to the answered column; second, write one real-life connection from your community; third, write one thing you are still curious about.' Set a 5-minute timer. Circulate — if a group is stuck on an 'unanswered' question, quietly read the question to them and ask: 'Did the written assessment today cover this?' After 5 minutes, call groups back. Go around all groups for 'Connections to Real Life' — insist on specificity ('a boat oar at Lake Victoria' is better than 'a lever'). After all groups share, hold up the original Lesson 1 driving question card. Ask the whole class: 'Answer this in one sentence — together.' Allow 10 seconds of murmuring, then cold-call one learner for the final sentence. Write it on the board below the driving question. Say: 'That sentence represents eight lessons of scientific thinking. Well done.'	learners and celebrates the intellectual journey of the entire sub-strand. The 'Remaining Wonder' notes validate scientific curiosity as an ongoing disposition, not a sign of incomplete learning. The real-life connections column directly links back to the driving question's framing of engineers and farmers in Kenya, ensuring the sub-strand ends with outward-facing, applied thinking.	accurate (i.e., learners correctly identify the pivot, effort, and how distance from pivot creates the moment)? (c) Do 'Remaining Wonder' notes reflect genuine extension thinking (e.g., 'What happens when the force is not perpendicular to the lever arm?') or surface-level confusion? The teacher uses this as final holistic formative data about the class's collective understanding of Sub-Strand 3.1.
Model Building Phase 📹 **VIDEO:** Equation with the variable in the denominator Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Linear, Exponential, and Quadratic Models: Bernoulli	Learners retrieve their cumulative model diagram from Lesson 1 (kept in their portfolios or science notebooks). They spend 6 minutes annotating and revising it to produce their FINAL MODEL. The final model must include: (1) a labelled diagram of the jembe showing pivot (blade-soil contact point), effort (hand on handle), and perpendicular distance; (2) the moment equation written on an arrow; (3) a brief note on anti-parallel forces (e.g., two hands on a steering wheel); (4) an annotation comparing their Lesson 1 model with their final model —	Distribute portfolios or ask learners to open to their Lesson 1 model page. Say: 'This is what you drew and wrote in Lesson 1. In the next 6 minutes, revise this model so it is complete. It must show: the jembe diagram with pivot, effort, and distance labelled; the moment equation; a note on couples; and your own annotation of what changed between Lesson 1 and today.' Set a 6-minute timer. Circulate — if a learner's final model still omits the perpendicular distance label, point to the arm of the diagram and ask: 'What is the name of this distance, and why	Final Model Revision with Lesson 1 Comparison — requiring learners to annotate their own Lesson 1 model (rather than draw a new one from scratch) makes conceptual change visceral and personal. Learners literally see, on the same page, where their understanding was incomplete and what it now contains. The reflection journal entry then externalises this metacognitive awareness in narrative form and pushes learners to bridge classroom learning to lived Kenyan contexts — fulfilling the original promise of the anchoring	Observable indicators in the final model: (a) pivot, effort, and perpendicular distance correctly labelled on the jembe diagram; (b) Moment = F × d written correctly; (c) Principle of Moments stated in words or symbols; (d) annotation describes at least one specific conceptual change from Lesson 1. Observable indicators in the reflection journal: learner names a specific real-life context (e.g., 'I will notice it when my mother stirs the ugali pot with a long wooden spoon' or 'when my father grips the jembe close to the blade'), and uses at least one piece of scientific

Effect (PLIX) Source: CK-12 🔍 Search ARES for similar readings	what was missing, what was wrong, what is now correct. Learners then write a Reflection Journal Entry (5 minutes, individual, in their own words) responding to the prompt: 'How has your understanding of turning effect of force changed since Lesson 1, and where will you notice it this week at home or on the farm?' Journals are submitted to the teacher at the end of the lesson.	does it appear in the formula?' WAIT 10 seconds. After 6 minutes, say: 'Now close the model and open a fresh page. Write the date and the heading: Reflection Journal — Turning Effect of Force. Answer this question in your own words — no copying from notes, no copying from a neighbour: How has your understanding of turning effect of force changed since Lesson 1, and where will you notice it this week at home or on the farm? You have 5 minutes.' Set timer. At end of 5 minutes: 'Write your name on both the model page and the journal page. Place them on the collection table as you leave.' End the lesson by reading the class's Final Explanation statement from the board aloud one more time. Say: 'You explained a puzzle that baffled you eight lessons ago. That is what scientists do. Well done.'	phenomenon.	vocabulary (moment, pivot, distance, force) naturally in their writing. Teacher reads all journals before the next class and provides written feedback within one week.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions and record your responses in your professional journal before the next teaching day:

1. ASSESSMENT VALIDITY: Did the unseen written problems (especially Item 3 on anti-parallel forces) genuinely differentiate between learners who understood the concepts and those who had only memorised procedures? If many learners scored well on Items 1–2 but poorly on Item 3, this signals that the treatment of couples across Lessons 4–6 needs strengthening in future iterations.

2. PRACTICAL ASSESSMENT: How fluently were learners able to verbalise the Principle of Moments during the metre-rule balancing task? If learners could balance the rule correctly but struggled to explain it in words, this suggests a gap between procedural competence and conceptual articulation — a target for future lesson design.

3. PHENOMENON CLOSURE: When the class constructed the three-sentence Final Explanation of the jembe puzzle, did it come quickly and confidently, or did it require significant teacher scaffolding? A smooth, learner-driven Final Explanation indicates the storyline was coherent. Heavy teacher intervention indicates the phenomenon was not revisited frequently enough across the eight lessons.

4. DQB QUALITY: Were the 'Connections to Real Life' notes on the DQB specific, accurate, and drawn from genuinely diverse Kenyan contexts (farm, fishing, kitchen, construction, transport)? Low diversity or inaccuracy suggests learners are not yet transferring understanding beyond the jembe example.

5. REFLECTION JOURNALS: What patterns do you notice in the reflection journal entries? Are learners identifying moments phenomena in genuinely personal, home-based contexts, or are they defaulting to textbook examples? Strong journals suggest the anchoring phenomenon successfully connected school science to lived experience. Weak journals suggest a need for more explicit 'science in daily life' prompts throughout the sub-strand, not only in the final lesson.

6. EQUITY CHECK: Were all learners — including those who struggled in early lessons — able to participate meaningfully in the practical assessment and the Final Explanation? If not, consider whether differentiated support (e.g., prompt cards, peer pairing) during the practical component would better serve all learners without compromising assessment integrity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed: (i) their own Lesson 1 prediction cards alongside their current understanding, making conceptual change visible; (ii) the metre-rule balancing task, where a physical system achieves rotational equilibrium only when masses are placed at distances that satisfy the Principle of Moments; (iii) the anchoring phenomenon image (jembe handle puzzle) displayed for the final time, now viewed through a fully developed scientific lens.
What did I learn?	Learners demonstrated that: the moment of a force equals the force multiplied by its perpendicular distance from the pivot ($M = F \times d$); a body is in rotational equilibrium when the sum of clockwise moments equals the sum of anti-clockwise moments; anti-parallel forces forming a couple produce a net turning effect equal to $F \times d$ with no resultant translational force; and the long-handled jembe is easier to use because the greater perpendicular distance from the pivot (blade-soil contact) produces a larger moment from the same applied force — fully explaining the anchoring phenomenon.
How does this explain the phenomenon?	Learners constructed a three-sentence class Final Explanation: 'A moment is the turning effect of a force, calculated as the force multiplied by its perpendicular distance from the pivot. In the jembe, the blade-soil contact is the pivot — a longer handle increases the perpendicular distance, so the same effort from the farmer creates a much larger moment, making it easier to break hard soil. This is why engineers and farmers design tools with longer handles or wider grips: to maximise the turning effect without increasing the effort required.'

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.